

Hospital Management System

HOSPITAL MANAGEMENT SYSTEM

A PROJECT SUBMITTED TO



BHAKTA KAVI NARSINH MEHTA UNIVERSITY

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COMPUTER SCIENCE

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1 Project Profile

1.1 Purpose

Hospital Management System (HMS) is a software application designed to automate and manage the day-to-day operations of a hospital. It integrates various departments such as registration, appointments, billing, pharmacy, laboratory, and patient records into a single centralized system. The system improves efficiency, reduces paperwork, minimizes errors, and ensures quick access to accurate patient and hospital data.

1.1.1 Objectives

- To computerize hospital administrative and clinical processes
- To maintain centralized and secure patient records
- To streamline appointment scheduling and doctor management
- To improve billing, pharmacy, and laboratory operations
- To enhance overall hospital efficiency and patient care

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1.2 Scope

The Hospital Management System(HMS) covers:

- Patient registration and record management
- Doctor and staff management
- Appointment scheduling
- Billing and payment processing
- Pharmacy and inventory management
- Laboratory test management
- Report generation and data analytics

The system can be used by hospitals, clinics, and diagnostic centers of any size.

1.2.1 Modules of the System

1. **Patient Management Module** – Registration, medical history, treatment details
2. **Doctor Management Module** – Doctor profiles, schedules, and specialization
3. **Appointment Module** – Online and offline appointment booking
4. **Billing Module** – Invoices, payments, and insurance details
5. **Pharmacy Module** – Medicine stock, sales, and expiry tracking
6. **Laboratory Module** – Test requests, results, and reports
7. **Admin Module** – User management, system settings, and reports

2 Objective

2.1 Technology Review

The Hospital Management System is built using modern web development technologies to ensure scalability, performance, and ease of maintenance. The backend is developed using Node.js, a powerful runtime environment that allows developers to run JavaScript on the server side.

Node.js is widely chosen for real-time applications due to its event-driven, non-blocking I/O model, which makes it lightweight and efficient. With Node.js, the system can handle multiple requests simultaneously, making it suitable for a hms management system where users perform frequent operations such as creating, updating, and deleting.

MongoDB is an open-source non-relational database management system widely used for storing, managing, and retrieving unstructured data efficiently. Instead of tables and rows, it stores data as JSON-like documents, making it easy to work with modern applications. MongoDB supports dynamic schemas, so fields can vary between documents.

MongoDB provides high performance, reliability, and scalability, along with features such as data integrity, transaction support, indexing, and security mechanisms. Its compatibility with multiple programming languages and operating systems enables seamless integration with web and software applications. Due to its

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flexibility, ease of use, and cost-effectiveness, MongoDB is commonly adopted in web development, data-driven systems, and cloud-based solutions.

The system primarily implements CRUD (Create, Read, Update, Delete) operations, which are the backbone of any data-driven application. CRUD ensures that users can efficiently manage their hms by adding new entries, retrieving existing data, modifying hms details, and removing completed or unnecessary records.

These operations are implemented using RESTful APIs in Node.js, providing a structured way to interact with the database. Additionally, frameworks and libraries such as Express.js are commonly integrated to simplify server-side development, improve routing, and manage middleware effectively.

To enhance usability, the system may incorporate front-end technologies like HTML, CSS, and JavaScript, or modern frameworks such as React or Angular, depending on the design requirements. This ensures a user-friendly interface where it can be managed seamlessly.

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2.2 Literature Review

HMS Hospital Management Systems have been widely studied and developed in both academic and industrial contexts. In today's fast-paced environment, effective hms organization is crucial for personal productivity as well as collaborative teamwork.

Traditional methods, such as paper-based planners and spreadsheets, often fail to provide real-time synchronization, accessibility, and scalability. As a result, digital solutions leveraging web and cloud technologies have gained popularity.

Research in hms management systems emphasizes the importance of intuitive interfaces, real-time synchronization, and flexible data handling. Several existing systems, such as Trello, Asana, and Microsoft To Do, highlight features like drag-and-drop hms organization, reminders, and team collaboration.

However, many of these platforms are either too complex for small-scale use or restricted by subscription-based models. This creates opportunities for building customized systems tailored to specific needs.

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Studies also indicate that integrating hms management systems with cloud-based databases improves accessibility and data consistency across devices. MongoDB's document-based storage model has been noted for its adaptability in managing unstructured or semi-structured data, which is common in hms management applications.

Node.js, due to its asynchronous nature, has been widely discussed in research for building real-time collaborative tools, including chat systems, project trackers, and managers.

By combining Node.js and MongoDB, Hms aligns with contemporary research and best practices in software engineering. It offers a balance of efficiency, scalability, and user-friendliness. The literature supports the use of CRUD-based architectures for managing core functionalities, ensuring that the users can perform essential hms operations reliably and efficiently with ease.

3 Requirement Analysis

3.1 Problem Definition

Hospitals are complex organizations that involve multiple departments such as registration, outpatient services, inpatient care, laboratories, pharmacies, billing, and administration.

In many hospitals, especially small and medium-sized healthcare institutions, these activities are still managed using manual or semi-computerized systems. This leads to inefficiencies, data redundancy, delays, errors in record keeping, and difficulty in accessing accurate patient information when needed.

The traditional manual hospital management process relies heavily on paper-based records for patient registration, medical history, prescriptions, laboratory reports, and billing details. Maintaining these records is time-consuming and prone to human errors such as misplacement of files, duplication of data, incorrect billing, and loss of critical medical information.

As the number of patients increases, managing appointments, tracking patient history, coordinating between departments, and generating reports becomes increasingly difficult. Another major issue is the lack of integration between hospital departments.

Doctors, laboratory staff, pharmacists, and billing personnel often work in isolation, resulting in poor communication and delays in patient care.

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For example, laboratory reports may not be available on time, prescriptions may be misread, and billing calculations may be inaccurate due to missing or inconsistent data. This negatively affects the quality of healthcare services and patient satisfaction.

Additionally, manual systems make it difficult to generate real-time reports for hospital management, such as bed availability, medicine stock levels, revenue details, and patient statistics. This limits effective decision-making, planning, and resource utilization. Security and confidentiality of patient data are also major concerns, as paper records can be easily accessed, damaged, or lost.

To overcome these problems, there is a need for an efficient, reliable, and integrated Hospital Management System. The proposed system aims to automate and manage all hospital-related activities through a centralized computerized platform. It will ensure accurate data storage, easy retrieval of patient information, seamless communication between departments, reduced workload, improved operational efficiency, and enhanced quality of patient care while maintaining data security and confidentiality.

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3.2 Process Model

A **process model** shows **how data and activities flow** through the hospital system—from a patient walking in to discharge and billing. It focuses on *who does what, in what order*, and *how information moves* between departments.

The **process model** of HMS can be divided into several key stages:

3.2.1 Patient Registration

Actors: Patient, Receptionist

Process:

- Patient provides personal details
- System generates **Patient ID**
- Records stored in patient database

Outputs: Patient record, registration confirmation

3.2.2 Appointment & OPD Management

Actors: Patient, Doctor, System

Process:

- Appointment scheduled
- Doctor views patient history
- Consultation details entered

Outputs: Diagnosis, prescriptions, test requests

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3.2.3 Laboratory & Diagnostic Services

Actors: Lab Technician, System

Process:

- Doctor orders lab tests
- Samples collected
- Test results uploaded to system

Outputs: Lab reports available to doctor & patient

3.2.4 Pharmacy Management

Actors: Pharmacist, System

Process:

- Doctor sends prescription
- Medicines issued
- Inventory updated automatically

Outputs: Medicine bill, stock update

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3.2.5 Billing & Payment

Actors: Billing Staff, Patient

Process:

- Charges calculated (consultation, lab, room, pharmacy)
- Invoice generated
- Payment processed (cash/card/insurance)

Outputs: Final bill, payment receipt

3.2.6 Discharge Process

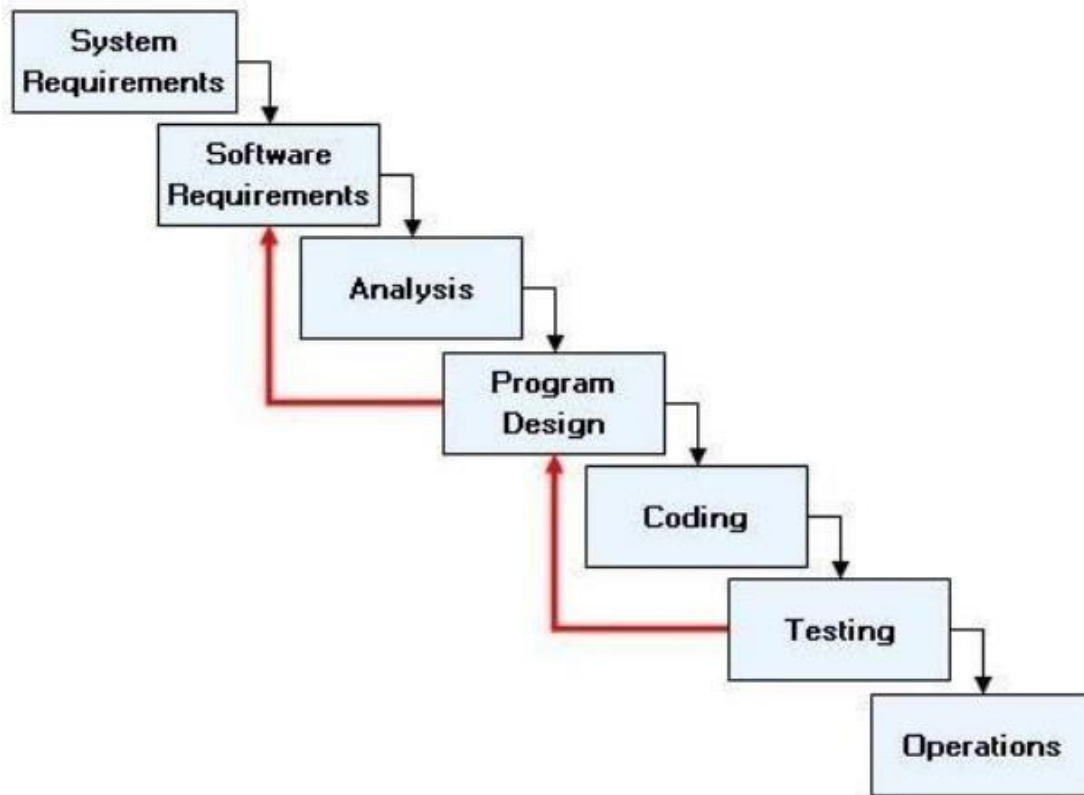
Actors: Doctor, Nurse, Billing Department

Process:

- Discharge summary prepared
- Final bill cleared
- Patient discharged

Outputs: Discharge summary, medical history archived

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3.3 Project Planning

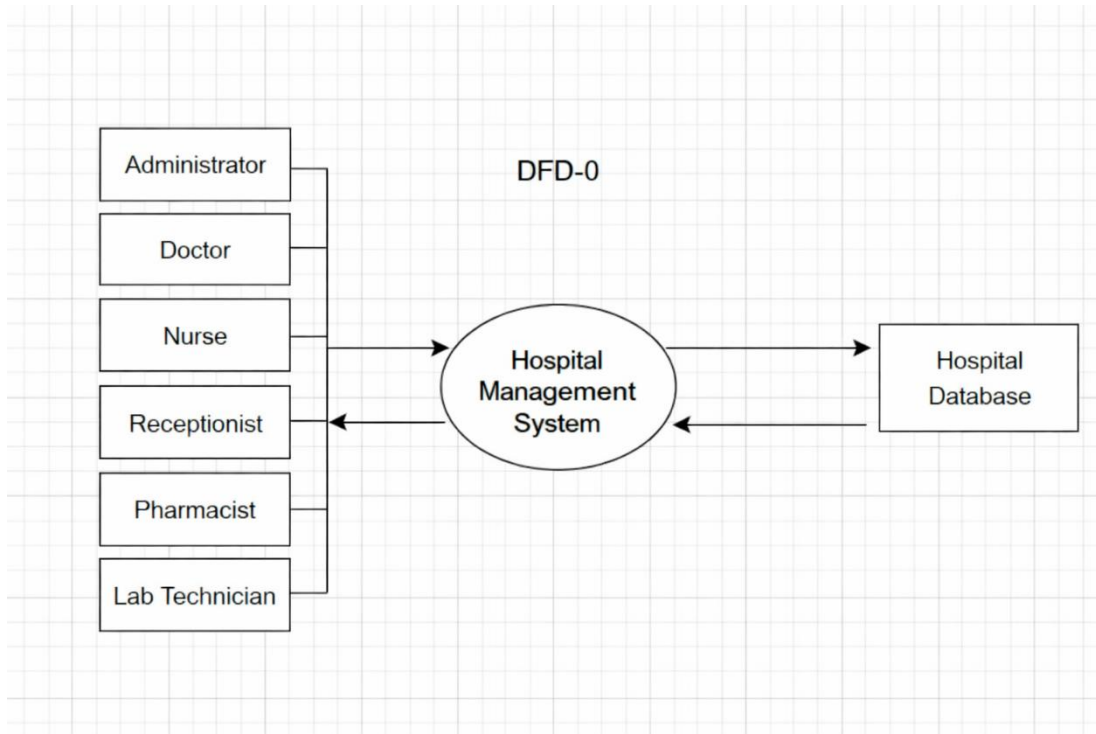
Phase	Timeline
Requirement Analysis	Week 1
Design & Mockups	Week 2
Development	Weeks 3 - 6
Testing	Weeks 7 - 8
Deployment	Week 9
Documentation	Week 10

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4 System Diagram

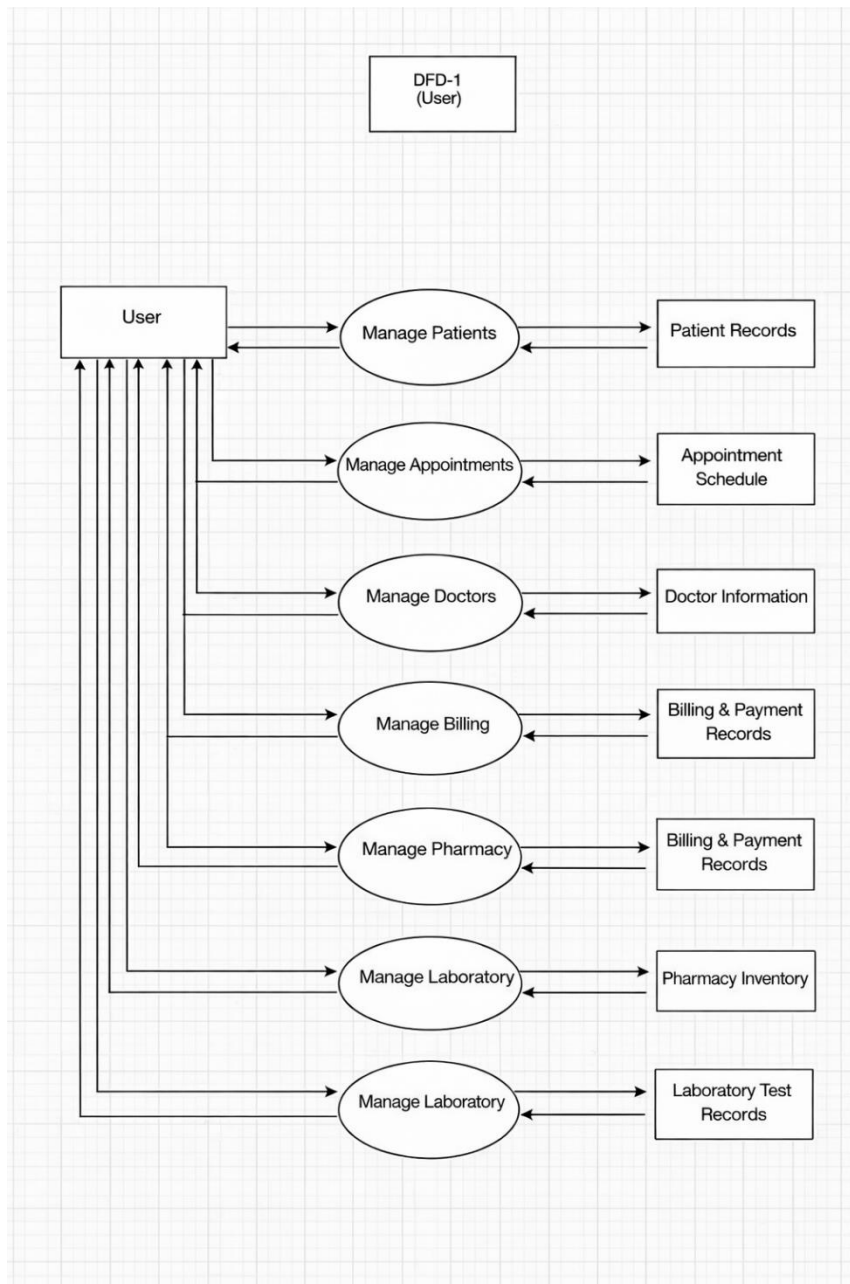
4.1 Data Flow Diagram

4.1.1 DFD-Level-0



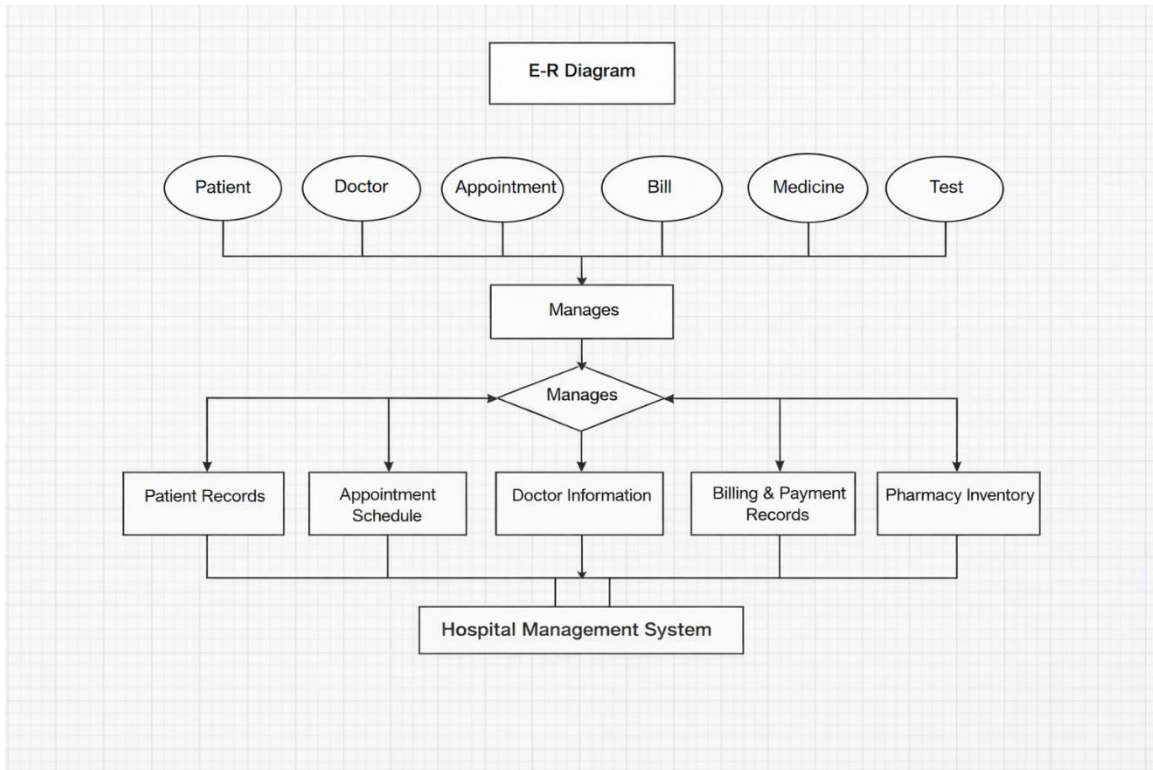
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4.1.2 DFD-Level-1



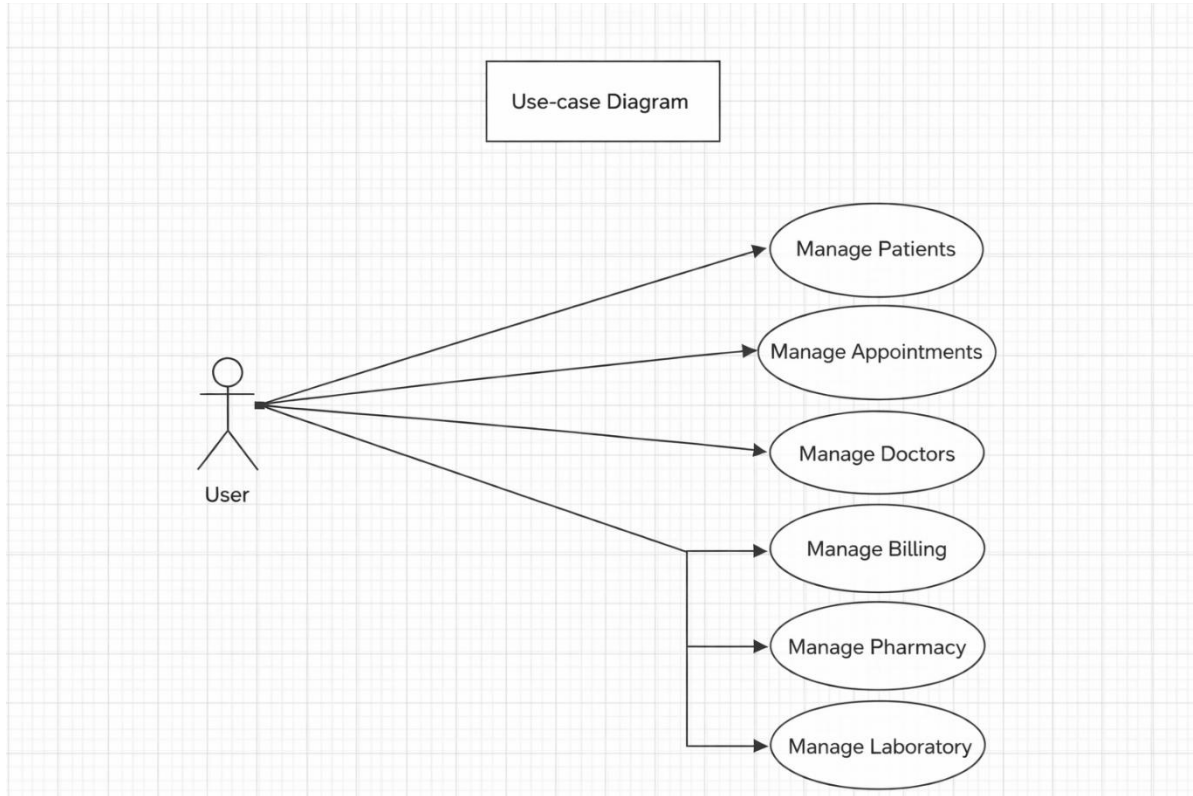
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4.2 E-R (Entity-Relationship) Diagram



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4.3 Use-Case Diagram



5 Data Dictionary

5.1 Tables

Table-1: Patient Table

Field	Data Type
patient_id	INT
first_name	VARCHAR
last_name	VARCHAR
gender	VARCHAR
date_of_birth	VARCHAR
phone	VARCHAR
address	VARCHAR

Table-2: Doctor Table

Field	Data Type
doctor_id	INT
doctor_name	VARCHAR
specialization	VARCHAR
phone	VARCHAR
email	VARCHAR
department	VARCHAR

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Table-3: Appointment Table

Field	Data Type
appointment_id	INT
patient_id	VARCHAR
doctor_id	VARCHAR
appointment_date	DATE
appointment_time	TIME
status	VARCHAR

Table-4: Billing Table

Field	Data Type
bill_id	INT
patient_id	INT
appointment_id	INT
total_amount	DECIMAL
payment_status	VARCHAR
payment_date	DATE

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Table-5: Pharmacy Table

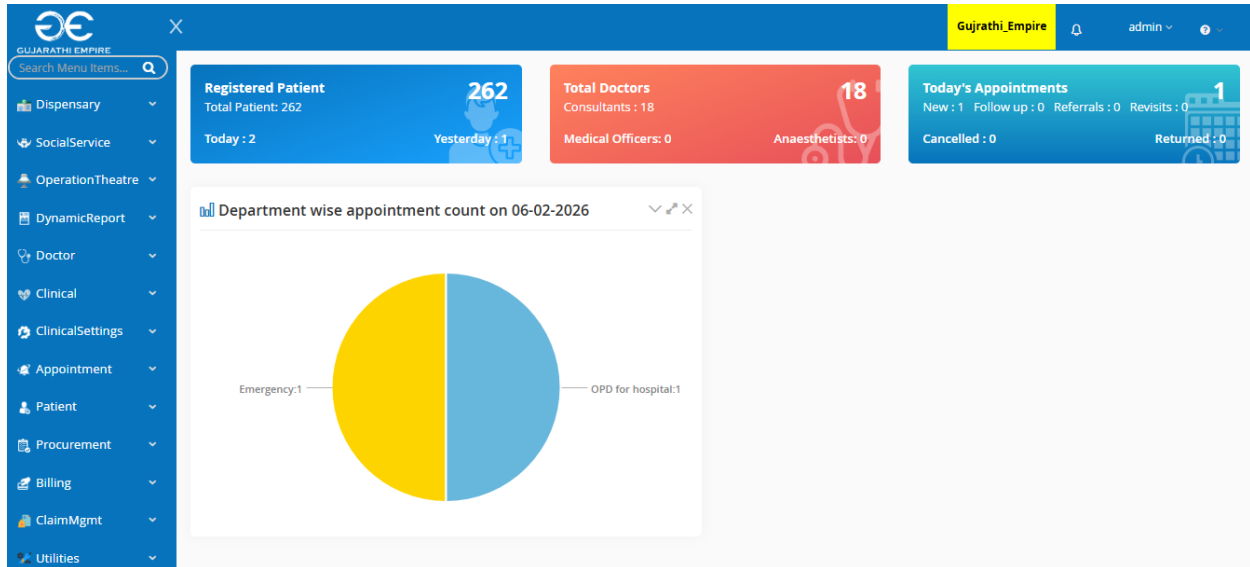
Field	Data Type
medicine_id	INT
medicine_name	VARCHAR
quantity	INT
price	DECIMAL
expiry_date	DATE

Table-6: Laboratory Test Table

Field	Data Type
test_id	INT
test_name	VARCHAR
test_cost	DECIMAL

6 User Interface (Frontend)

6.1 Dashboard



- The Dashboard is the main landing page of the Hospital Management System after user login. It provides a quick overview of hospital activities and helps users access important modules efficiently.
- A dashboard is a visual user interface that provides a centralized, real-time, or near real-time, overview of key performance indicators (KPIs), metrics, and critical data points from various sources.

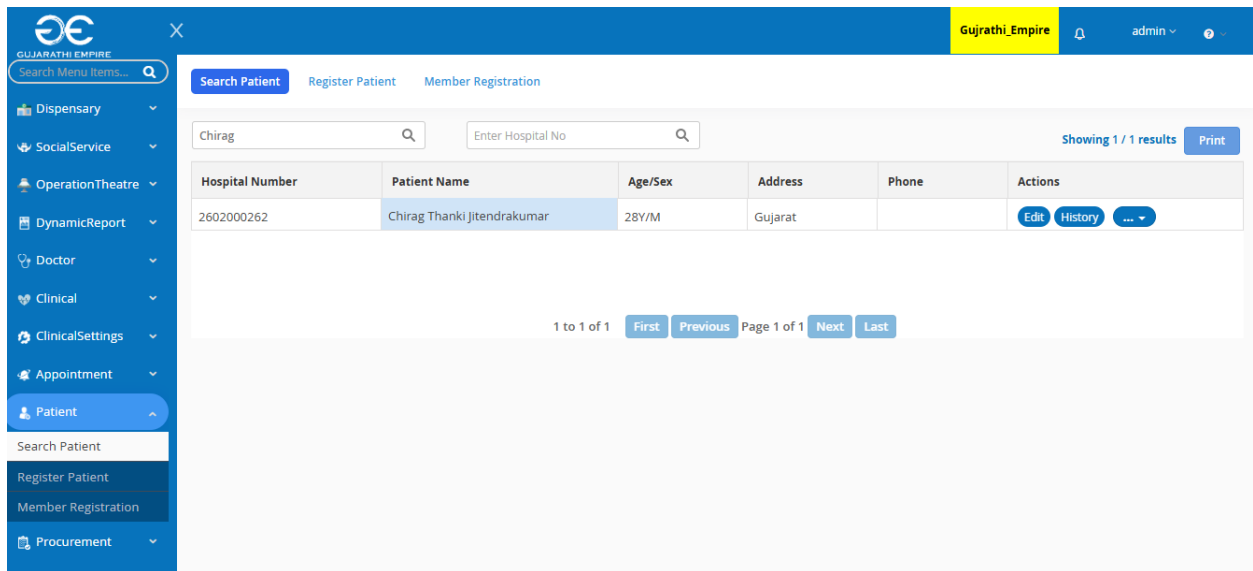
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6.2 Patient Registration

The screenshot displays the 'Gujrathi_Empire' Hospital Management System interface. The top navigation bar includes the system logo, a search menu, and user information (admin). The left sidebar lists various modules: Dispensary, SocialService, OperationTheatre, DynamicReport, Doctor, Clinical, ClinicalSettings, Appointment, Patient (selected), Search Patient, Register Patient, Member Registration, and Procurement. The main content area is titled 'Register Patient' and features tabs for 'Basic Information', 'Address', 'Guarantor', 'Insurance', and 'Kin/Emergency Contact'. The 'Basic Information' tab is active, showing a form with fields for Salutation, Gender, Caste, Marital status, Notifications, Employer Info, Previous Last Name, Occupation, Email, Race, First Name, Middle Name, Last Name, Date Of Birth, Age, Phone number, LandLine Number, and Country. The 'First Name' field is highlighted with a blue border. A 'Register Patient' button is located in the top right corner of the form area.

- Patient Registration is the first and most important module of the Hospital Management System. It is used to record and store patient details digitally when a patient visits the hospital for the first time.
- Patient Registration is the module that records patient details, generates a unique patient ID, and stores information digitally for future medical reference.

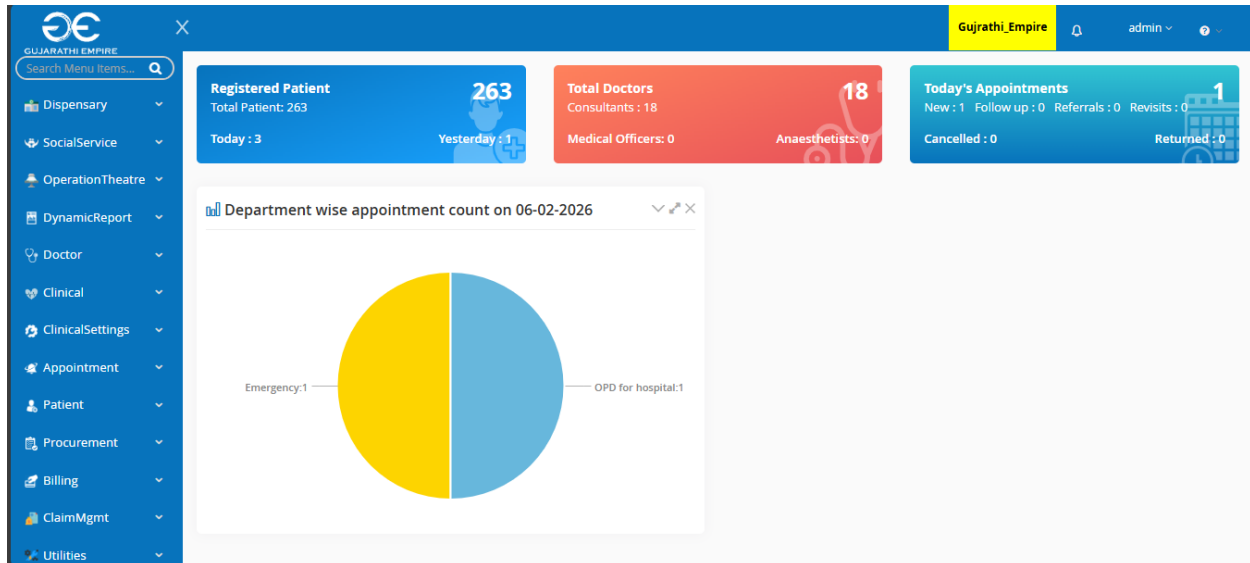
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- Features of Patient Registration Module
 1. Fast and user-friendly interface
 2. Avoids duplicate patient records
 3. Supports search and update of patient details
 4. Ensures data accuracy and security
 5. Role-based access control

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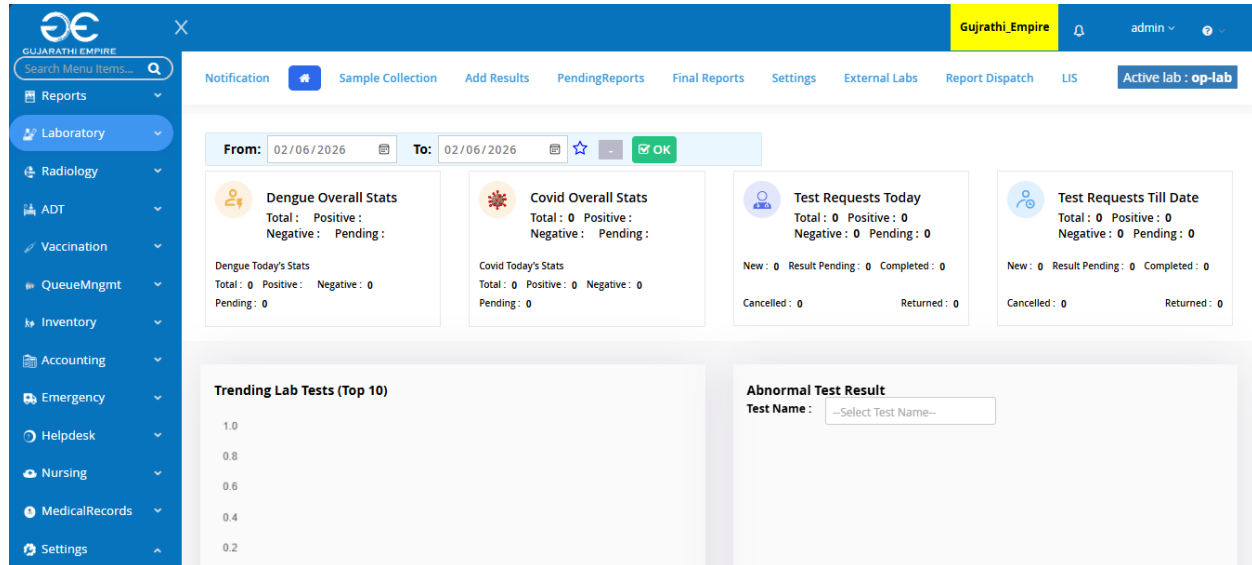
6.3 Dashboard Menus



- Dashboard menus shows different menus in this Hospital Management System (HMS) like:
 1. Dispensary
 2. Operation Theatre
 3. Doctor
 4. Clinical
 5. Appointment
 6. Patient
 7. Billing
 8. Procurement
 9. Claim Management
 10. Inventory Management

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6.4 Laboratory



- This tab shows different laboratory related services like
 1. Dengue Statistics
 2. Test Requests
 3. Doctor orders lab tests
 4. Samples collected
 5. Test results uploaded to system

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6.5 Departments

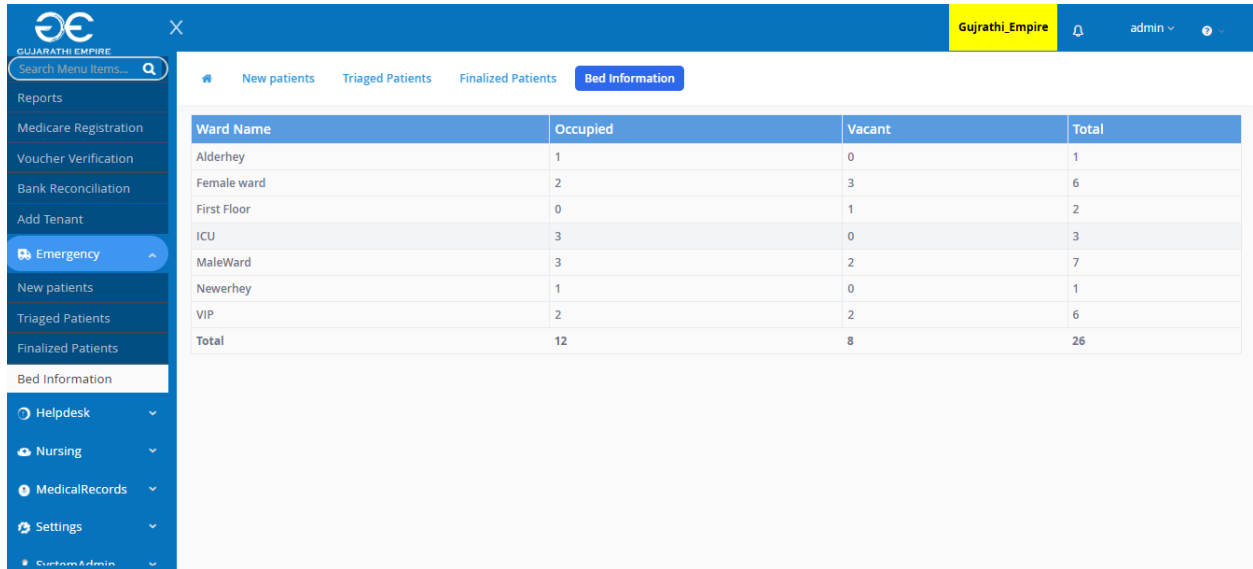
The screenshot displays the 'Departments' section of the Hospital Management System. The interface includes a top navigation bar with the system name 'Gujrathi_Empire' and a user profile 'admin'. A left sidebar contains various menu items, with 'Departments' highlighted. The main content area features a 'Manage Department' tab, a '+Add Department' button, and a search bar. Below the search bar is a table listing departments with columns for Code, Name, Parent Department, Description, Is Active, Is Appointment Ap..., and Action. The table shows 31 results.

Code	Name	Parent Department	Description	Is Active	Is Appointment Ap...	Action
123	123	Administration	123	true	true	Edit
ADM	Administration		Administration Department	true	true	Edit
ADM	Administrator	Administration	Administrator	true	true	Edit
AD	alnamer	Pathology		true	false	Edit
BILL	Billing	sample dep		true	true	Edit
BLOO_DEP1	BLOO DEPART			true	true	Edit
DPT-001	Department One			true	false	Edit
DEPT_SAC	DEPT_SAC			true	true	Edit

- Shows different departments like:
 1. Radiology
 2. Admit & Discharge Transfer (ADT)
 3. Security
 4. Employee
 5. Billing
 6. Clinical

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6.6 Bed Information



Ward Name	Occupied	Vacant	Total
Alderhey	1	0	1
Female ward	2	3	6
First Floor	0	1	2
ICU	3	0	3
MaleWard	3	2	7
Newerhey	1	0	1
VIP	2	2	6
Total	12	8	26

- Shows Information Regarding Beds:
 1. Bed Availability
 2. Ward detail
 3. No. of beds occupied
 4. No. of beds vacant
 5. Total no. of beds

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6.7 Employee Information

Employee Name	Designation	Department	Phone No.	Ext.	Spee...	Room...	Office Hours
Dr PHRM Anonymous Doctor	FullTime	Administration	0				0
Dr. Alexandro Berg		Administration					0
Dr. Shiv .	FullTime	OPD for hospital					0
Dr. Shirish .	Doctor	OPD for hospital					0
Mr. Harish Name	Lab Technicain	Pathology					0
Mr. Ramesh .		Radiology					0
Dr. Suresh D	Doctor	InPatientBedCha...					0
Dr. Ramesh D		Emergency					0
Mr. ahmed dom	FullTime	ophthmlogy					0
Mr. kam no more	FullTime	Administration		911	922	66	8
Dr. CLK		Administration					n

- The Employee Information tab is used to store, manage, and maintain information of all hospital staff in the system.
- It helps the administration manage employees efficiently and ensures proper role-based access

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6.8 OPD - Outpatient Department Registration

The screenshot displays the 'Gujarathi_Empire' Hospital Management System interface. The top navigation bar includes 'Gujarathi_Empire', a user profile 'admin', and a search bar. The left sidebar lists various modules: Appointment, Patient, Procurement, Purchase Request, Purchase Order, Goods Arrival Notification, Quotation, Settings, Reports, Billing (highlighted), ClaimMgmt, Utilities, and MktReferral. The main content area is titled 'Search Patients' and features a '+Add New Patient' button. Below this, there are two search sections: 'Search OPD Patients:' and 'Search IPD Patients:'. The 'Search OPD Patients:' section shows a search result for 'Chirag Thanki J' with the following details: Name: Chirag Thanki J, Hospital #: 2602000262, Age/Sex: 24Y/ Male, Visit: Outpatient, and Address: Gujarat. Below the patient details, there are five action buttons: 'New Billing Request', 'Add Deposit', 'Show Bill History' (with a checkbox), 'Patient Sticker', and 'Add Diagnosis'.

- OPD (Outpatient Department) Information manages details of patients who visit the hospital for consultation and treatment without being admitted.
- This module records each outpatient visit and supports smooth doctor–patient interaction.

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The screenshot displays the 'Booking OT Schedule | New Patient' modal in a Hospital Management System. The modal is overlaid on a dashboard with a sidebar menu on the left containing options like 'Dispensary', 'SocialService', 'OperationTh', 'BookingList', 'Settings', 'Reports', 'DynamicReport', 'Doctor', 'Clinical', 'ClinicalSettings', 'Appointment', 'Patient', 'Search Patient', and 'Register Patient'. The modal itself has a title bar with a close button. Inside, there are two fields: 'Select Visit Type*' with radio buttons for 'IP' (selected) and 'OP/ER', and 'Select Patient*' with a text input containing 'Chirag Thanki'. A 'Load' button is on the right. Below the modal, a table with columns 'Ho...', 'Patient N...', 'A', 'I.', 'OT Sc...', 'E', 'Surge...', 'P', 'S', 'BillingItems', 'Su...', 'S.', 'OT Co...', 'O', 'R', and 'Actions' is visible. The table is currently empty, showing 'No Rows To Show'. At the bottom, there is a pagination control showing '0 to 0 of 0' and buttons for 'First', 'Previous', 'Page 0 of 0', 'Next', and 'Last'.

- Gives information about Operation Theatre booking and patient details like name and reason for the operation.
- OPD Information manages outpatient visits by recording consultation details, diagnosis, prescriptions, and lab tests without patient admission.

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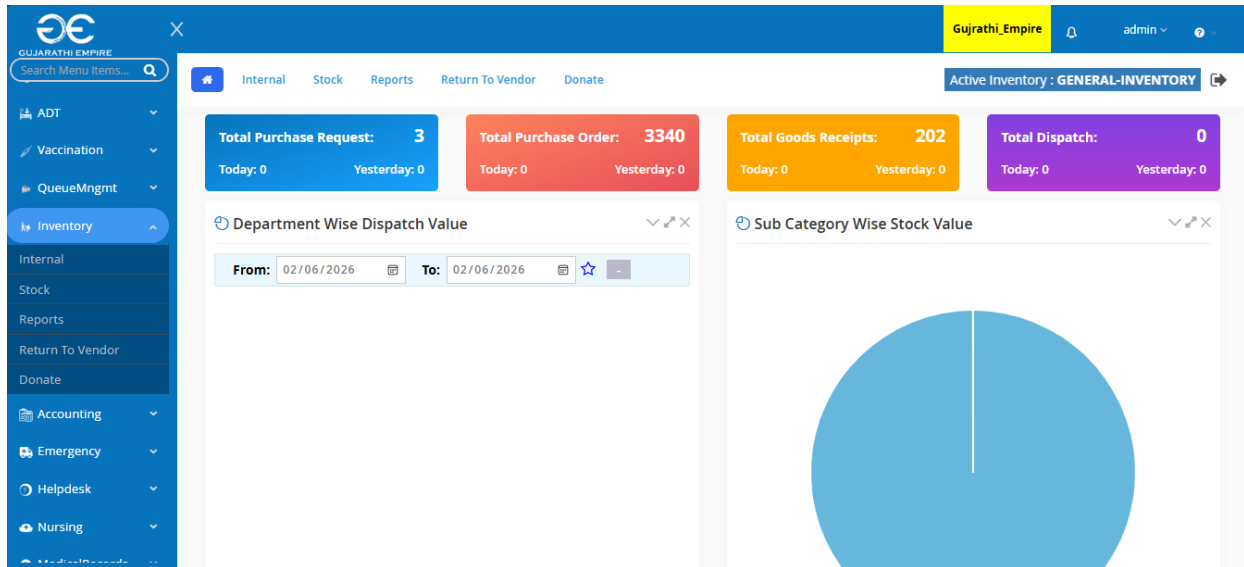
6.9 Patient Insurance

The screenshot displays the 'Patient Insurance' module within the 'GUJARATHI EMPIRE' Hospital Management System. The interface features a blue sidebar on the left with a search bar and a menu including 'Dispensary', 'SocialService', 'OperationTheatre', 'DynamicReport', 'Doctor', 'Clinical', 'ClinicalSettings', 'Appointment', 'Patient' (highlighted), 'Search Patient', 'Register Patient', 'Member Registration', and 'Procurement'. The top header is blue with the system logo, a search bar, and user information 'Gujrathi_Empire' and 'admin'. The main content area has a light blue background with tabs for 'Basic Information', 'Address', 'Guarantor', 'Insurance' (selected), and 'Kin/Emergency Contact'. A green 'Register Patient' button is in the top right. The 'Insurance' tab contains the following fields: 'Insurance Provider*' (dropdown), 'Insurance Name' (text), 'Card Number [NTSCNo]*' (text), 'Insurance Number [NSSINO]*' (text), 'IMISCode*' (text), and 'Initial Balance*' (text with value '0'). An 'Add Insurance' button is located below these fields.

- The Patient Insurance module is used to record, manage, and process insurance details of patients who are covered under health insurance schemes.
- It helps the hospital handle cashless treatment and insurance claims efficiently.
- Store patient insurance information securely
- Simplify billing and claim processing
- Support cashless and insured treatments

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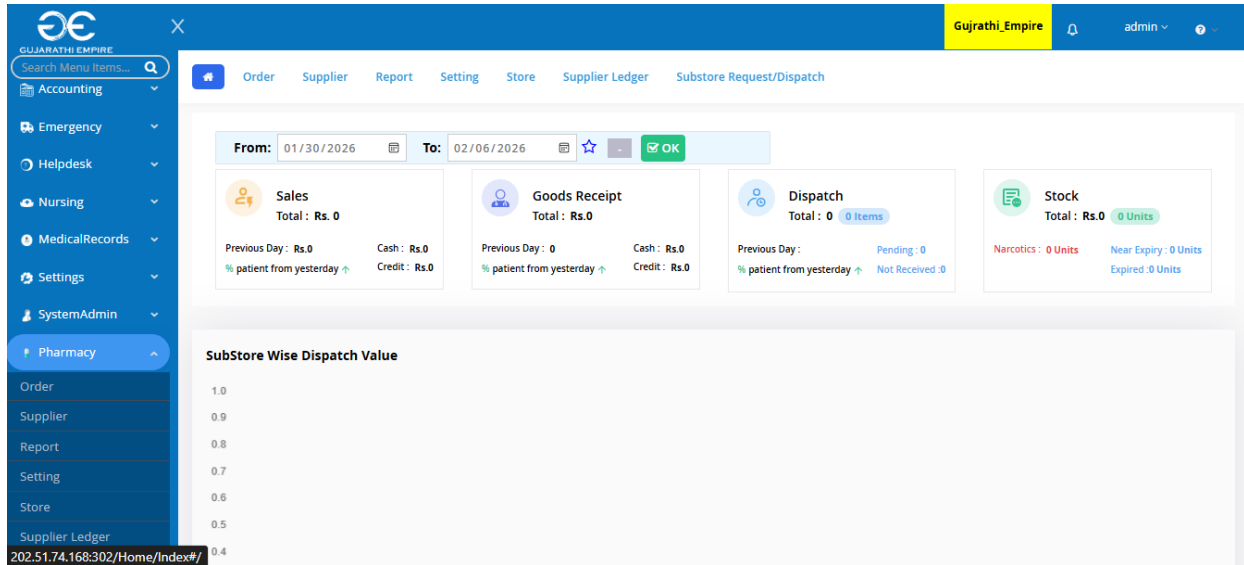
6.10 Inventory Management



- The Inventory Management module is used to track, control, and manage hospital stock such as medicines, medical equipment, surgical items, and consumables.
- It ensures that required items are always available and prevents shortages or wastage.
- Inventory Management tracks hospital stock, monitors usage and expiry, and ensures timely availability of medical supplies.

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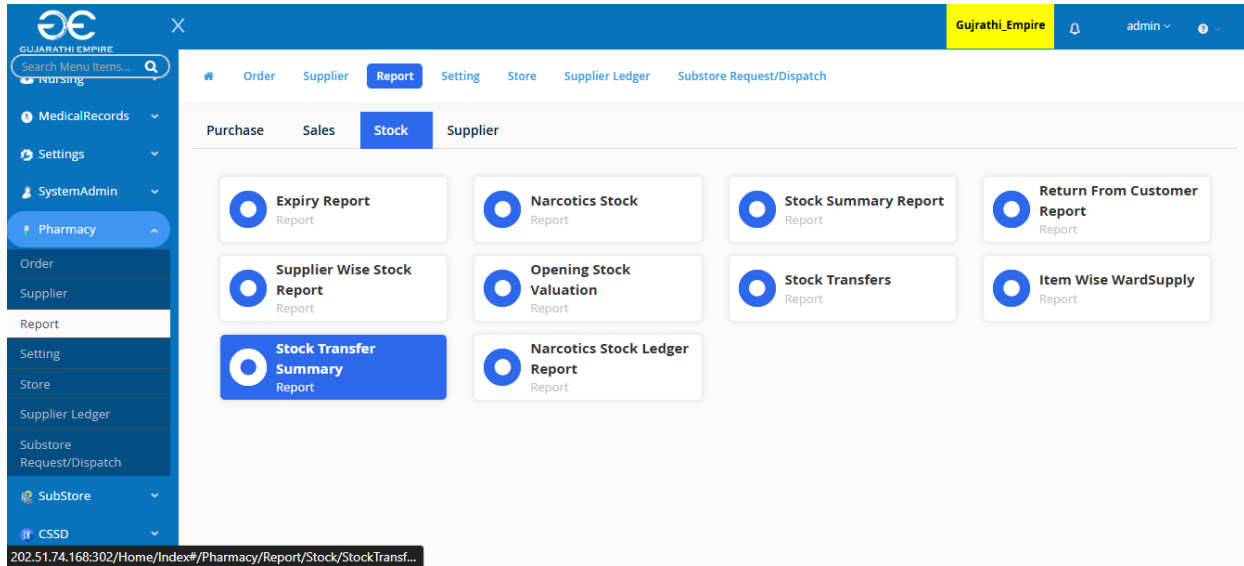
6.11 Pharmacy Support



- The Pharmacy Support module manages the dispensing of medicines, maintains medicine inventory, and ensures accurate billing based on doctor prescriptions.
- It plays a vital role in patient treatment and hospital operations.

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6.12 Stock Report



- The Stock Report provides a detailed summary of all inventory items available in the hospital, especially medicines and medical supplies.
- It helps administrators and pharmacy staff monitor stock levels and make timely purchase decisions.

7 Server-Side (Backend)

7.1 Server.js

```
const express = require("express");

const mongoose = require("mongoose");

const bodyParser = require("body-parser");

const Hms = require("../models/Hms");

const app = express();

app.use(bodyParser.urlencoded({ extended: true }));

app.set("view engine", "ejs");

app.use(express.urlencoded({ extended: true }));

mongoose.connect("mongodb://127.0.0.1:27017/hms", {
  useNewUrlParser: true,
  useUnifiedTopology: true,
});
```

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// Routes

```
app.get("/", async (req, res) => {  
    const hms = await hms.find();  
    res.render("index", { hms });  
});
```

```
app.post("/add", async (req, res) => {  
    const new hms = new hms({ hms: req.body.hms });  
    await new hms.save();  
    res.redirect("/");  
});
```

```
app.post("/update/:id", async (req, res) => {  
    const { hms } = req.body;  
    await Hms.findByIdAndUpdate(req.params.id, { hms: hms });  
    res.redirect("/");  
});
```

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```
app.post("/delete/:id", async (req, res) => {  
    await Hms.findByIdAndDelete(req.params.id);  
    res.redirect("/");  
});  
  
app.post("/toggle/:id", async (req, res) => {  
    const hms = await Hms.findById(req.params.id);  
    hms.completed = !hms.completed;  
    await hms.save();  
    res.redirect("/");  
});
```

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7.2 app.js:

```
(function () {  
    'use strict';  
  
    var isOn = window.location.hostname === 'blueimp.io',  
        url = isOn ? '//jquery-file-upload.appspot.com/' : 'server/js';  
  
    angular.module('demo', [  
        'blueimp.fileupload'  
    ])  
        .config([  
            '$httpProvider', 'fileUploadProvider',  
            function ($httpProvider, fileUploadProvider) {  
                delete $httpProvider.defaults.headers.common['X-Requested-With'];  
                fileUploadProvider.defaults.redirect = window.location.href.replace(  
                    /\^[^V]*$/,  
                    '/cors/result.html?%s'  
                );  
                if (isOn) {  
                    // Demo settings:  
                    angular.extend(fileUploadProvider.defaults, {  
                        // Enable image resizing, except for Android and Opera,  
                        // which actually support image resizing, but fail to  
                        // send Blob objects via XHR requests:  
                        disableImageResize: /Android(?!.*Chrome)|Opera/  
                    },  
                    .test(window.navigator.userAgent),
```

Hospital Management System

```
        maxFileSize: 999000,
        acceptFileTypes: /(\\.|\\V)(gif|jpe?g|png)$/i
    });
}
}
])

.controller('DemoFileUploadController', [
    '$scope', '$http', '$filter', '$window',
    function ($scope, $http) {
        $scope.options = {
            url: url
        };

        if (!isOn) {
            $scope.loadingFiles = true;
            $http.get(url)
                .then(
                    function (response) {
                        $scope.loadingFiles = false;
                        $scope.queue = response.data.files || [];
                    },
```

Hospital Management System

```
        function () {
            $scope.loadingFiles = false;
        }
    );

}

}

])

.controller('FileDestroyController', [
    '$scope', '$http',
    function ($scope, $http) {
        var file = $scope.file,
            state;

        if (file.url) {
            file.$state = function () {
                return state;
            };

            file.$destroy = function () {
                state = 'pending';
                return $http({
                    url: file.deleteUrl,
                    method: file.deleteType
```

Hospital Management System

```
}).then(  
    function () {  
        state = 'resolved';  
        $scope.clear(file);  
    },  
    function () {  
        state = 'rejected';  
    }  
);  
};  
  
} else if (!file.$cancel && !file._index) {  
    file.$cancel = function () {  
        $scope.clear(file);  
    };  
};
```

Hospital Management System

7.3 login.js:

```
var usrJson = { user: 'chirag', pwd: 'chirag' };
```

```
$(document).ready(function () {  
  
    $('#login').show().animate({ opacity: 1 }, 2000);  
  
    $('.logo').show().animate({ opacity: 1, top: '40%' }, 800, function ()  
  
    {  
  
        $('.logo').show().delay(1200).animate({ opacity: 1, top: '12%' }, 300, function  
        ()  
  
        {  
  
            $('.formLogin').animate({ opacity: 1, left: '0' }, 300);  
  
            $('.userbox').animate({ opacity: 0 }, 200).hide();  
  
        });  
  
    })  
  
    $(".on_off_checkbox").iphoneStyle();  
  
    $('.tip a ').tipsy({ gravity: 'sw' });  
  
    $('.tip input').tipsy({ trigger: 'focus', gravity: 'w' });  
  
});
```

Hospital Management System

```
$('.userload').click(function (e) {  
    $('.formLogin').animate({ opacity: 1, left: '0' }, 300);  
    $('.userbox').animate({ opacity: 0 }, 200, function () {  
        $('.userbox').hide();  
    });  
});  
  
$('#but_logins').click(function (e) {  
  
    if (document.formLogin.username.value == "" ||  
document.formLogin.password.value == "") {  
        showError("Please Input Username / Password");  
        $('.inner').jrumble({ x: 4, y: 0, rotation: 0 });  
        $('.inner').trigger('startRumble');  
        setTimeout($(".inner").trigger("stopRumble"), 500);  
        setTimeout('hideTop()', 5000);  
        return false;  
    }  
}
```

Hospital Management System

```
else {  
    var curUsr = document.formLogin.username.value;  
    var curPwd = document.formLogin.password.value;  
    return true;  
  
    if (curUsr != usrJson.user || curPwd != usrJson.pwd) {  
        showError("Invalid Username / Password");  
        $('.inner').jrumble({ x: 4, y: 0, rotation: 0 });  
        $('.inner').trigger('startRumble');  
        setTimeout($('.inner').trigger("stopRumble"), 500);  
        setTimeout('hideTop()', 5000);  
        return false;  
    }  
}  
  
hideTop();  
loading('Checking', 1);  
setTimeout("unloading()", 2000);  
setTimeout("Login()", 2500);  
});
```

Hospital Management System

```
function Login()
{
    $("#login").animate({ opacity: 1, top: '49%' }, 200, function () {
        $('.userbox').show().animate({ opacity: 1 }, 500);
        $("#login").animate({ opacity: 0, top: '60%' }, 500, function () {
            $(this).fadeOut(200, function () {
                $(".text_success").slideDown();
                $("#successLogin").animate({ opacity: 1, height: "200px" }, 500);
            });
        })
    })
    setTimeout("window.location.href='dashboard.html'", 3000);
}

$('#alertMessage').click(function () {
    hideTop();
});
```

Hospital Management System

```
function showError(str)
{
    $('#alertMessage').addClass('error').html(str).stop(true, true).show().animate({
opacity: 1, right: '10' }, 500);

    if (document.formLogin.username.value == "")
    {
        $('#username_id').addClass('text-error').html(str).stop(true,
true).show().animate({ opacity: 1, right: '10' }, 500);
    }

    if (document.formLogin.password.value == "") {
        $('#password').addClass('text-error').html(str).stop(true,
true).show().animate({ opacity: 1, right: '10' }, 500);
    }
}

function showSuccess(str) {
    $('#alertMessage').removeClass('error').html(str).stop(true,
true).show().animate({ opacity: 1, right: '10' }, 500);

    $('#username_id').removeClass('text-error').html(str).stop(true,
true).show().animate({ opacity: 1, right: '10' }, 500);

    $('#password').removeClass('text-error').html(str).stop(true,
true).show().animate({ opacity: 1, right: '10' }, 500);
}
```

Hospital Management System

```
function hideTop() {  
    $('#alertMessage').animate({ opacity: 0, right: '-20' }, 500, function () {  
        $(this).hide(); });  
}  
  
function loading(name, overlay) {  
    $('body').append('<div id="overlay"></div><div id="preloader">' + name +  
        '..</div>');  
  
    if (overlay == 1) {  
        $('#overlay').css('opacity', 0.1).fadeIn(function () { $('#preloader').fadeIn();  
    });  
  
        return false;  
    }  
  
    $('#preloader').fadeIn();  
}  
  
function unloading() {  
    $('#preloader').fadeOut('fast', function () { $('#overlay').fadeOut(); });  
}
```

Hospital Management System

7.4 index.html:

```
<!DOCTYPE html>

<html>

<head>

  <title>Chirag Thanki</title>

  <link rel="shortcut icon" href="/views/favicon.png" type="image/x-icon">

  <style>

    body {

      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;

      max-width: 700px;

      margin: 40px auto;

      padding: 20px;

      background: #f4f7fa;

      color: #333;

    }

    h1,

    h2 {

      text-align: center;

      margin-bottom: 20px;

      color: #2c3e50;}
```

Hospital Management System

```
form[action="/add"] {
```

```
    display: flex;
```

```
    justify-content: center;
```

```
    margin-bottom: 25px;
```

```
}
```

```
input[type="text"] {
```

```
    flex: 1;
```

```
    padding: 10px 15px;
```

```
    border: 2px solid #ddd;
```

```
    border-radius: 8px 0 0 8px;
```

```
    font-size: 16px;
```

```
    outline: none;
```

```
    transition: border 0.3s;
```

```
}
```

```
input[type="text"]:focus {
```

```
    border-color: #3498db;
```

```
}
```

Hospital Management System

```
button {  
  
    padding: 10px 15px;  
  
    border: none;  
  
    background: #3498db;  
  
    color: white;  
  
    font-size: 15px;  
  
    border-radius: 0 8px 0;  
  
    cursor: pointer;  
  
    transition: background 0.3s, transform 0.2s;  
  
}
```

```
button:hover {  
  
    background: #2980b9;  
  
    transform: translateY(-2px);  
  
}
```

```
ul {  
  
    list-style: none;  
  
    padding: 0;  
  
}
```

Hospital Management System

```
li {  
  
    display: flex;  
  
    align-items: center;  
  
    justify-content: space-between;  
  
    background: white;  
  
    padding: 12px 16px;  
  
    margin-bottom: 12px;  
  
    border-radius: 10px;  
  
    box-shadow: 0 2px 5px rgba(0, 0, 0, 0.08);  
  
    transition: transform 0.2s;  
  
}
```

```
li:hover {  
  
    transform: scale(1.02);  
  
}
```

```
.completed {  
  
    text-decoration: line-through;  
  
    color: #888;  
  
}
```

Hospital Management System

```
li form {
```

```
    display: inline;
```

```
}
```

```
li button {
```

```
    background: #27ae60;
```

```
    border-radius: 6px;
```

```
    margin-left: 8px;
```

```
    padding: 6px 12px;
```

```
    font-size: 14px;
```

```
}
```

```
li button:hover {
```

```
    background: #1e8449;
```

```
}
```

```
li form[action*="delete"] button {
```

```
    background: #e74c3c;
```

```
}
```

Hospital Management System

```
li form[action*="delete"] button:hover {
```

```
    background: #c0392b;
```

```
}
```

```
li form[action*="toggle"] button {
```

```
    background: #f39c12;
```

```
}
```

```
li form[action*="toggle"] button:hover {
```

```
    background: #d68910;
```

```
}
```

```
span {
```

```
    flex: 1;
```

```
    margin: 0 10px;
```

```
    font-size: 16px;
```

```
}
```

```
a {
```

```
    display: inline-block;
```

```
    font-size: 16px;
```

```
    color: #3498db;
```

```
    transition: all 0.3s ease-in-out;
```

```
}
```

Hospital Management System

```
a:hover {  
    transform: scale(1.1);  
    color: #e67e22;  
    text-shadow: 0 0 8px rgba(230, 126, 34, 0.7);  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>HMS</h1>
```

```
<h2>Hospital Management System</h2>
```

```
<form action="/add" method="POST">
```

```
<input type="text" name="hms" placeholder="New Hms" required />
```

```
<button type="submit">Add</button>
```

```
</form>
```

```
<ul>
```

```
<% hms.forEach(hms=> { %>
```

Hospital Management System

```
<li>

    <!-- Toggle -->

    <form action="/toggle/<%= hms._id %>" method="POST"
style="display:inline;">

        <button type="submit">

            <%= hms.completed ? "Undo" : "Done" %>

        </button>

    </form>

    <!-- Hms text -->

    <span id="text-<%= hms._id %>" class="<%= hms.completed ?
'completed' : " %>">

        <%= hms.hms %>

    </span>

    <!-- Hidden update form -->

    <!-- <form id="form-<%= hms._id %>" action="/update/<%= hms._id
%>" method="POST" style="display:none;">

        <input type="text" name="hms" value="<%= hms.hms %>" required
/>

        <button type="submit">Save</button>
```

Hospital Management System

```
<button type="button" onclick="cancelEdit('<%= hms._id  
%>')">Cancel</button>
```

```
</form> -->
```

```
<!-- Edit + Delete -->
```

```
<!-- <button type="button" onclick="editHms('<%= hms._id  
%>')">Edit</button> -->
```

```
<form action="/delete/<%= hms._id %>" method="POST"  
style="display:inline;">
```

```
<button type="submit">Delete</button>
```

```
</form>
```

```
</li>
```

```
<% }) %>
```

```
</ul>
```

Hospital Management System

```
<script>

    function editHms(id) {

        document.getElementById("text-" + id).style.display = "none";

        document.getElementById("form-" + id).style.display = "inline";

    }

    function cancelEdit(id) {

        document.getElementById("text-" + id).style.display = "inline";

        document.getElementById("form-" + id).style.display = "none";

    }

</script>

</body>

</html>
```

Hospital Management System

7.5 package.json

```
{
  "name": "hospital-management-system",
  "version": "1.0.0",
  "description": "Hospital Management System using Node.js and MongoDB",
  "main": "server.js",
  "scripts": {
    "start": "node server.js",
    "dev": "nodemon server.js"
  },
  "author": "Your Name",
  "license": "MIT",
  "dependencies": {
    "bcryptjs": "^2.4.3",
    "body-parser": "^1.20.2",
    "cors": "^2.8.5",
    "dotenv": "^16.3.1",
    "express": "^4.18.2",
    "jsonwebtoken": "^9.0.2",
    "mongoose": "^8.0.3",
    "morgan": "^1.10.0"
  },
  "devDependencies": {
    "nodemon": "^3.0.2"
  }
}
```

Hospital Management System

7.6 inbox.js

```
var AppInbox = function () {  
  
    var content = $('#inbox-content');  
    var listListing = "";  
  
    var loadInbox = function (el, name) {  
        var url = 'app_inbox_inbox.html';  
        var title = el.attr('data-title');  
        listListing = name;  
  
        App.blockUI({  
            target: content,  
            overlayColor: 'none',  
            animate: true  
        });  
  
        toggleButton(el);  
  
        $.ajax({  
            type: "GET",  
            cache: false,  
            url: url,  
            dataType: "html",  
            success: function(res)
```

Hospital Management System

```
{
  toggleButton(el);

  App.unblockUI('.inbox-content');

  $('inbox-nav > li.active').removeClass('active');
  el.closest('li').addClass('active');
  $('inbox-header > h1').text(title);

  content.html(res);

  if (Layout.fixContentHeight) {
    Layout.fixContentHeight();
  }
},
error: function(xhr, ajaxOptions, thrownError)
{
  toggleButton(el);
},
async: false
});

// handle group checkbox:
jQuery('body').on('change', '.mail-group-checkbox', function () {
  var set = jQuery('.mail-checkbox');
```

Hospital Management System

```
var checked = jQuery(this).is(":checked");
jQuery(set).each(function () {
    $(this).attr("checked", checked);
});
});
}

var loadMessage = function (el, name, resetMenu) {
    var url = 'app_inbox_view.html';

    App.blockUI({
        target: content,
        overlayColor: 'none',
        animate: true
    });

    toggleButton(el);

    var message_id = el.parent('tr').attr("data-messageid");

    $.ajax({
        type: "GET",
        cache: false,
        url: url,
        dataType: "html",
```

Hospital Management System

```
data: {'message_id': message_id},
success: function(res)
{
    App.unblockUI(content);

    toggleButton(el);

    if (resetMenu) {
        $('.inbox-nav > li.active').removeClass('active');
    }
    $('.inbox-header > h1').text('View Message');

    content.html(res);
    Layout.fixContentHeight();
},
error: function(xhr, ajaxOptions, thrownError)
{
    toggleButton(el);
},
async: false
});
}

var initWysihtml5 = function () {
    $('.inbox-wysihtml5').wysihtml5({
```

Hospital Management System

```
"stylesheets": ["../assets/global/plugins/bootstrap-wysihtml5/wysiwyg-color.css"]
```

```
});
```

```
}
```

```
var initFileupload = function () {
```

```
$('#fileupload').fileupload({
```

```
    // Uncomment the following to send cross-domain cookies:
```

```
    //xhrFields: {withCredentials: true},
```

```
    url: '../assets/global/plugins/jquery-file-upload/server/php/',
```

```
    autoUpload: true
```

```
});
```

```
// Upload server status check for browsers with CORS support:
```

```
if ($.support.cors) {
```

```
    $.ajax({
```

```
        url: '../assets/global/plugins/jquery-file-upload/server/php/',
```

```
        type: 'HEAD'
```

```
    }).fail(function () {
```

```
        $('
```

```
            .text('Upload server currently unavailable - ' +
```

```
            new Date())
```

```
            .appendTo('#fileupload');
```

```
    });
```

```
}
```

Hospital Management System

```
}
```

```
var loadCompose = function (el) {
```

```
    var url = 'app_inbox_compose.html';
```

```
    App.blockUI({
```

```
        target: content,
```

```
        overlayColor: 'none',
```

```
        animate: true
```

```
    });
```

```
    toggleButton(el);
```

```
    // load the form via ajax
```

```
    $.ajax({
```

```
        type: "GET",
```

```
        cache: false,
```

```
        url: url,
```

```
        dataType: "html",
```

```
        success: function(res)
```

```
        {
```

```
            App.unblockUI(content);
```

```
            toggleButton(el);
```

```
            $('inbox-nav > li.active').removeClass('active');
```

Hospital Management System

```
$('.inbox-header > h1').text('Compose');

content.html(res);

initFileupload();
initWysihtml5();

$('.inbox-wysihtml5').focus();
Layout.fixContentHeight();
},
error: function(xhr, ajaxOptions, thrownError)
{
    toggleButton(el);
},
async: false
});
}

var loadReply = function (el) {
    var messageid = $(el).attr("data-messageid");
    var url = 'app_inbox_reply.html';

    App.blockUI({
        target: content,
        overlayColor: 'none',
```

Hospital Management System

```
        animate: true
    });

toggleButton(el);

// load the form via ajax
$.ajax({
    type: "GET",
    cache: false,
    url: url,
    dataType: "html",
    success: function(res)
    {
        App.unblockUI(content);
        toggleButton(el);

        $('.inbox-nav > li.active').removeClass('active');
        $('.inbox-header > h1').text('Reply');

        content.html(res);
        $('[name="message"]').val($('#reply_email_content_body').html());

        handleCCInput(); // init "CC" input field

        initFileupload();
```

Hospital Management System

```
initWysihtml5();
Layout.fixContentHeight();
},
error: function(xhr, ajaxOptions, thrownError)
{
    toggleButton(el);
},
async: false
});
}
```

```
var handleCCInput = function () {
    var the = $('inbox-compose .mail-to .inbox-cc');
    var input = $('inbox-compose .input-cc');
    the.hide();
    input.show();
    $('close', input).click(function () {
        input.hide();
        the.show();
    });
}
```

```
var handleBCCInput = function () {

    var the = $('inbox-compose .mail-to .inbox-bcc');
```

Hospital Management System

```
var input = $('inbox-compose .input-bcc');  
the.hide();  
input.show();  
$('.close', input).click(function () {  
    input.hide();  
    the.show();  
});  
}
```

```
var toggleButton = function(el) {  
    if (typeof el == 'undefined') {  
        return;  
    }  
    if (el.attr("disabled")) {  
        el.attr("disabled", false);  
    } else {  
        el.attr("disabled", true);  
    }  
}
```

```
return {  
    //main function to initiate the module  
    init: function () {  
  
        // handle compose btn click
```

Hospital Management System

```
$('.inbox').on('click', '.compose-btn', function () {  
    loadCompose($(this));  
});  
  
// handle discard btn  
$('.inbox').on('click', '.inbox-discard-btn', function(e) {  
    e.preventDefault();  
    loadInbox($(this), listListing);  
});  
  
// handle reply and forward button click  
$('.inbox').on('click', '.reply-btn', function () {  
    loadReply($(this));  
});  
  
// handle view message  
$('.inbox').on('click', '.view-message', function () {  
    loadMessage($(this));  
});  
  
// handle inbox listing  
$('.inbox-nav > li > a').click(function () {  
    loadInbox($(this), 'inbox');  
});
```

Hospital Management System

```
//handle compose/reply cc input toggle
$('.inbox-content').on('click', '.mail-to .inbox-cc', function () {
    handleCCInput();
});

//handle compose/reply bcc input toggle
$('.inbox-content').on('click', '.mail-to .inbox-bcc', function () {
    handleBCCInput();
});

//handle loading content based on URL parameter
if (App.getURLParameter("a") === "view") {
    loadMessage();
} else if (App.getURLParameter("a") === "compose") {
    loadCompose();
} else {
    $('.inbox-nav > li:first > a').click();
}
}
};
})();

jQuery(document).ready(function() {
    AppInbox.init();
});
```

Hospital Management System

7.7 dashboard.js

```
var Dashboard = function() {  
    return {  
        initJQVMAP: function() {  
            if (!jQuery().vectorMap) {  
                return;  
            }  
  
            var showMap = function(name) {  
                jQuery('.vmaps').hide();  
                jQuery('#vmap_' + name).show();  
            }  
  
            var setMap = function(name) {  
                var map = jQuery('#vmap_' + name);  
  
                if (map.size() !== 1) {  
                    return;  
                }  
  
                var data = {  
                    map: 'world_en',  
                    backgroundColor: null,  
                    borderColor: '#333333',  
                    borderOpacity: 0.5,
```

Hospital Management System

```
borderWidth: 1,
color: '#c6c6c6',
enableZoom: true,
hoverColor: '#c9dfaf',
hoverOpacity: null,
values: sample_data,
normalizeFunction: 'linear',
scaleColors: ['#b6da93', '#909cae'],
selectedColor: '#c9dfaf',
selectedRegion: null,
showTooltip: true,
onLabelShow: function(event, label, code) {

},
onRegionOver: function(event, code) {
    if (code == 'ca') {
        event.preventDefault();
    }
},
onRegionClick: function(element, code, region) {
    var message = 'You clicked "' + region + '" which has the code: ' +
code.toUpperCase();
    alert(message);
}
};
```

Hospital Management System

```
data.map = name + '_en';
```

```
map.width(map.parent().parent().width());
```

```
map.show();
```

```
map.vectorMap(data);
```

```
map.hide();
```

```
}
```

```
setMap("world");
```

```
setMap("usa");
```

```
setMap("europe");
```

```
setMap("russia");
```

```
setMap("germany");
```

```
showMap("world");
```

```
jQuery('#regional_stat_world').click(function() {
```

```
    showMap("world");
```

```
});
```

```
jQuery('#regional_stat_usa').click(function() {
```

```
    showMap("usa");
```

```
});
```

```
jQuery('#regional_stat_europe').click(function() {
```

```
    showMap("europe");
```

Hospital Management System

```
});  
jQuery('#regional_stat_russia').click(function() {  
    showMap("russia");  
});  
jQuery('#regional_stat_germany').click(function() {  
    showMap("germany");  
});  
  
$('#region_statistics_loading').hide();  
$('#region_statistics_content').show();  
  
App.addResizeHandler(function() {  
    jQuery('.vmaps').each(function() {  
        var map = jQuery(this);  
        map.width(map.parent().width());  
    });  
});  
},  
  
initCalendar: function() {  
    if (!jQuery().fullCalendar) {  
        return;  
    }  
  
    var date = new Date();
```

Hospital Management System

```
var d = date.getDate();
var m = date.getMonth();
var y = date.getFullYear();

var h = {};

if ($('#calendar').width() <= 400) {
    $('#calendar').addClass("mobile");
    h = {
        left: 'title, prev, next',
        center: "",
        right: 'today,month,agendaWeek,agendaDay'
    };
} else {
    $('#calendar').removeClass("mobile");
    if (App.isRTL()) {
        h = {
            right: 'title',
            center: "",
            left: 'prev,next,today,month,agendaWeek,agendaDay'
        };
    } else {
        h = {
            left: 'title',
            center: "",
```

Hospital Management System

```
        right: 'prev,next,today,month,agendaWeek,agendaDay'
    };
}
}
```

```
$('#calendar').fullCalendar('destroy'); // destroy the calendar
```

```
$('#calendar').fullCalendar({ //re-initialize the calendar
```

```
    disableDragging: false,
```

```
    header: h,
```

```
    editable: true,
```

```
    events: [{
```

```
        title: 'All Day',
```

```
        start: new Date(y, m, 1),
```

```
        backgroundColor: App.getBrandColor('yellow')
```

```
    }, {
```

```
        title: 'Long Event',
```

```
        start: new Date(y, m, d - 5),
```

```
        end: new Date(y, m, d - 2),
```

```
        backgroundColor: App.getBrandColor('blue')
```

```
    }, {
```

```
        title: 'Repeating Event',
```

```
        start: new Date(y, m, d - 3, 16, 0),
```

```
        allDay: false,
```

```
        backgroundColor: App.getBrandColor('red')
```

```
    }, {
```

Hospital Management System

```
        title: 'Repeating Event',
        start: new Date(y, m, d + 6, 16, 0),
        allDay: false,
        backgroundColor: App.getBrandColor('green')
    }, {
        title: 'Meeting',
        start: new Date(y, m, d + 9, 10, 30),
        allDay: false
    }, {
        allDay: false
    }, {
        title: 'Click for Google',
        start: new Date(y, m, 28),
        end: new Date(y, m, 29),
        backgroundColor: App.getBrandColor('yellow'),
        url: 'http://google.com/'
    }
    ]
    });
},

initCharts: function() {
    if (!jQuery.plot) {
        return;
    }
}
```

Hospital Management System

```
function showChartTooltip(x, y, xValue, yValue) {  
    $('<div id="tooltip" class="chart-tooltip">' + yValue + '</div>').css({  
        position: 'absolute',  
        display: 'none',  
        top: y - 40,  
        left: x - 40,  
        border: '0px solid #ccc',  
        padding: '2px 6px',  
        'background-color': '#fff'  
    }).appendTo("body").fadeIn(200);  
}
```

```
var data = [];
```

```
var totalPoints = 250;
```

```
// random data generator for plot charts
```

```
function getRandomData() {  
    if (data.length > 0) data = data.slice(1);  
    // do a random walk  
    while (data.length < totalPoints) {  
        var prev = data.length > 0 ? data[data.length - 1] : 50;  
        var y = prev + Math.random() * 10 - 5;  
        if (y < 0) y = 0;  
        if (y > 100) y = 100;
```

Hospital Management System

```
        data.push(y);
    }
    // zip the generated y values with the x values
    var res = [];
    for (var i = 0; i < data.length; ++i) res.push([i, data[i]])
    return res;
}

function randValue() {
    return (Math.floor(Math.random() * (1 + 50 - 20))) + 10;
}

if ($('#site_statistics').size() != 0) {

    $('#site_statistics_loading').hide();
    $('#site_statistics_content').show();

    var plot_statistics = $.plot($("#site_statistics"), [{
        data: visitors,
        lines: {
            fill: 0.6,
            lineWidth: 0
        },
        grid: {
            hoverable: true,
            clickable: true,
```

Hospital Management System

```
        tickColor: "#eee",
        borderColor: "#eee",
        borderWidth: 1
    }
});
var previousPoint = null;
$("#site_statistics").bind("plothover", function(event, pos, item) {
    $("#x").text(pos.x.toFixed(2));
    $("#y").text(pos.y.toFixed(2));
    if (item) {
        if (previousPoint != item.dataIndex) {
            previousPoint = item.dataIndex;

            $("#tooltip").remove();
            var x = item.datapoint[0].toFixed(2),
                y = item.datapoint[1].toFixed(2);

            showChartTooltip(item.pageX, item.pageY, item.datapoint[0],
item.datapoint[1] + ' visits');
        }
    } else {
        $("#tooltip").remove();
        previousPoint = null;
    }
});
}
```

Hospital Management System

```
if ($('#site_activities').size() != 0) {  
    //site activities  
    var previousPoint2 = null;  
    $('#site_activities_loading').hide();  
    $('#site_activities_content').show();  
  
    var data1 = [  
        ['DEC', 300],  
        ['JAN', 600],  
        ['FEB', 1100],  
        ['MAR', 1200],  
        ['APR', 860],  
        ['MAY', 1200],  
        ['JUN', 1450],  
        ['JUL', 1800],  
        ['AUG', 1200],  
        ['SEP', 600]  
    ];  
    var plot_statistics = $.plot($("#site_activities"),  
        [{  
            data: data1,  
            lines: {  
                fill: 0.2,  
                lineWidth: 0,  
            },  
        }],
```

Hospital Management System

```
    color: ['#BAD9F5']
  }, {
    data: data1,
    points: {
      show: true,
      fill: true,
      radius: 4,
      fillColor: "#9ACAE6",
      lineWidth: 2
    },
    color: '#9ACAE6',
    shadowSize: 1
  }, {
    data: data1,
    lines: {
      show: true,
      fill: false,
      lineWidth: 3
    },
    color: '#9ACAE6',
    shadowSize: 0
  }],

  {
```

Hospital Management System

```
xaxis: {
  tickLength: 0,
  tickDecimals: 0,
  mode: "categories",
  min: 0,
  font: {
    lineHeight: 18,
    style: "normal",
    variant: "small-caps",
    color: "#6F7B8A"
  }
},
yaxis: {
  ticks: 5,
  tickDecimals: 0,
  tickColor: "#eee",
  font: {
    lineHeight: 14,
    style: "normal",
    variant: "small-caps",
    color: "#6F7B8A"
  }
},
grid: {
  hoverable: true,
```

Hospital Management System

```
        clickable: true,  
        tickColor: "#eee",  
        borderColor: "#eee",  
        borderWidth: 1  
    }  
});
```

```
var time = new Date();  
var time_str = (time.getHours() + ':' + time.getMinutes());  
var tpl = "  
    tpl += '<li class=\"out\">';  
    tpl += '<img class=\"avatar\" alt=\"\" src=\"\" + Layout.getLayoutImgPath() +  
'avatar1.jpg\"/>';  
    tpl += '<div class=\"message\">';  
    tpl += '<span class=\"arrow\"></span>';  
    tpl += '<a href=\"#\" class=\"name\">Bob Nilson</a>&nbsp;';  
    tpl += '<span class=\"datetime\">at ' + time_str + '</span>';  
    tpl += '<span class=\"body\">';  
    tpl += text;  
    tpl += '</span>';  
    tpl += '</div>';  
    tpl += '</li>';  
  
var msg = list.append(tpl);
```

Hospital Management System

```
input.val("");
```

```
var getLastPostPos = function() {  
    var height = 0;  
    cont.find("li.out, li.in").each(function() {  
        height = height + $(this).outerHeight();  
    });
```

```
    return height;  
}
```

```
cont.find('.scroller').slimScroll({  
    scrollTo: getLastPostPos()  
});  
}
```

```
$('#body').on('click', '.message .name', function(e) {  
    e.preventDefault(); // prevent click event
```

```
    var name = $(this).text(); // get clicked user's full name  
    input.val('@' + name + ':'); // set it into the input field  
    App.scrollTo(input); // scroll to input if needed  
});
```

```
btn.click(handleClick);
```

Hospital Management System

```
input.keypress(function(e) {  
    if (e.which == 13) {  
        handleClick(e);  
        return false; //<---- Add this line  
    }  
});  
},
```

```
initDashboardDaterange: function() {  
    if (!jQuery().daterangepicker) {  
        return;  
    }  
}
```

```
$('#dashboard-report-range').daterangepicker({  
    "ranges": {  
        'Today': [moment(), moment()],  
        'Yesterday': [moment().subtract('days', 1), moment().subtract('days',  
1)],  
        'Last 7 Days': [moment().subtract('days', 6), moment()],  
        'Last 30 Days': [moment().subtract('days', 29), moment()],  
        'This Month': [moment().startOf('month'), moment().endOf('month')],  
        'Last Month': [moment().subtract('month', 1).startOf('month'),  
moment().subtract('month', 1).endOf('month')]  
    },  
    "locale": {
```

Hospital Management System

```
"format": "MM/DD/YYYY",  
"separator": " - ",  
"applyLabel": "Apply",  
"cancelLabel": "Cancel",  
"fromLabel": "From",  
"toLabel": "To",  
"customRangeLabel": "Custom",  
"daysOfWeek": [  
    "Su",  
    "Mo",  
    "Tu",  
    "We",  
    "Th",  
    "Fr",  
    "Sa"  
],  
"monthNames": [  
    "January",  
    "February",  
    "March",  
    "April",  
    "May",  
    "June",  
    "July",  
    "August",
```

Hospital Management System

```
    "September",  
    "October",  
    "November",  
    "December"  
  ],  
  "firstDay": 1  
},
```

```
initAmChart1: function() {  
  if (typeof(AmCharts) === 'undefined' || $('#dashboard_amchart_1').size()  
=== 0) {  
    return;  
  }  
}
```

```
var chartData = [{  
  "date": "2026-01-05",  
  "distance": 480,  
  "townName": "Miami",  
  "townName2": "Miami",  
  "townSize": 10,  
  "latitude": 25.83,  
  "duration": 501  
}, {  
  "date": "2012-01-06",  
  "distance": 386,  
  "townName": "Tallahassee",
```

Hospital Management System

```
"townSize": 7,  
"latitude": 30.46,  
"duration": 443  
, {  
  "date": "2012-01-07",  
  "distance": 348,  
  "townName": "New Orleans",  
  "townSize": 10,  
  "latitude": 29.94,  
  "duration": 405  
, {  
  "date": "2012-01-08",  
  "distance": 238,  
  "townName": "Houston",  
  "townName2": "Houston",  
  "townSize": 16,  
  "latitude": 29.76,  
  "duration": 309  
, {  
  "date": "2012-01-09",  
  "distance": 218,  
  "townName": "Dalas",  
  "townSize": 17,  
  "latitude": 32.8,  
  "duration": 287
```

Hospital Management System

```
}, {  
  "date": "2012-01-10",  
  "distance": 349,  
  "townName": "Oklahoma City",  
  "townSize": 11,  
  "latitude": 35.49,  
  "duration": 485  
}, {  
  "date": "2012-01-11",  
  "distance": 603,  
  "townName": "Kansas City",  
  "townSize": 10,  
  "latitude": 39.1,  
  "duration": 890  
}, {  
  "date": "2012-01-12",  
  "distance": 534,  
  "townName": "Denver",  
  "townName2": "Denver",  
  "townSize": 18,  
  "latitude": 39.74,  
  "duration": 810  
}, {  
  "date": "2012-01-13",  
  "townName": "Salt Lake City",
```

Hospital Management System

```
"townSize": 12,  
"distance": 425,  
"duration": 670,  
"latitude": 40.75,  
"alpha": 0.4  
}, {  
  "date": "2012-01-14",  
  "latitude": 36.1,  
  "duration": 470,  
  "townName": "Las Vegas",  
  "townName2": "Las Vegas",  
  "bulletClass": "lastBullet"  
}, {  
  "date": "2012-01-15"  
}];  
  
var chart = AmCharts.makeChart("dashboard_amchart_1", {  
  type: "serial",  
  fontSize: 12,  
  fontFamily: "Open Sans",  
  dataDateFormat: "YYYY-MM-DD",  
  dataProvider: chartData,  
  
  addClassNames: true,  
  startDuration: 1,  
  color: "#6c7b88",
```

Hospital Management System

```
marginLeft: 0,

categoryField: "date",
categoryAxis: {
  parseDates: true,
  minPeriod: "DD",
  autoGridCount: false,
  gridCount: 50,
  gridAlpha: 0.1,
  gridColor: "#FFFFFF",
  axisColor: "#555555",
  dateFormats: [{
    period: 'DD',
    format: 'DD'
  }, {
    period: 'WW',
    format: 'MMM DD'
  }, {
    period: 'MM',
    format: 'MMM'
  }, {
    period: 'YYYY',
    format: 'YYYY'
  }]
},
```

Hospital Management System

```
valueAxes: [{  
    id: "a1",  
    title: "distance",  
    gridAlpha: 0,  
    axisAlpha: 0  
}, {  
    id: "a2",  
    position: "right",  
    gridAlpha: 0,  
    axisAlpha: 0,  
    labelsEnabled: false  
}, {  
    id: "a3",  
    title: "duration",  
    position: "right",  
    gridAlpha: 0,  
    axisAlpha: 0,  
    inside: true,  
    duration: "mm",  
    durationUnits: {  
        DD: "d. ",  
        hh: "h ",  
        mm: "min",  
        ss: ""
```

Hospital Management System

```
    }  
  }],  
  graphs: [{  
    id: "g1",  
    valueField: "distance",  
    title: "distance",  
    type: "column",  
    fillAlphas: 0.7,  
    valueAxis: "a1",  
    balloonText: "[[value]] miles",  
    legendValueText: "[[value]] mi",  
    legendPeriodValueText: "total: [[value.sum]] mi",  
    lineColor: "#08a3cc",  
    alphaField: "alpha",  
  }, {  
    id: "g2",  
    valueField: "latitude",  
    classNameField: "bulletClass",  
    title: "latitude/city",  
    type: "line",  
    valueAxis: "a2",  
    lineColor: "#786c56",  
    lineThickness: 1,  
    legendValueText: "[[description]]/[[[value]]]",  
    descriptionField: "townName",
```

Hospital Management System

```
bullet: "round",
bulletSizeField: "townSize",
bulletBorderColor: "#02617a",
bulletBorderAlpha: 1,
bulletBorderThickness: 2,
bulletColor: "#89c4f4",
labelText: "[[townName2]]",
labelPosition: "right",
balloonText: "latitude:[[value]]",
showBalloon: true,
animationPlayed: true,
}, {
  id: "g3",
  title: "duration",
  valueField: "duration",
  type: "line",
  valueAxis: "a3",
  lineAlpha: 0.8,
  lineColor: "#e26a6a",
  balloonText: "[[value]]",
  lineThickness: 1,
  legendValueText: "[[value]]",
  bullet: "square",
  bulletBorderColor: "#e26a6a",
  bulletBorderThickness: 1,
```

Hospital Management System

```
bulletBorderAlpha: 0.8,
dashLengthField: "dashLength",
animationPlayed: true
}],

chartCursor: {
    zoomable: false,
    categoryBalloonDateFormat: "DD",
    cursorAlpha: 0,
    categoryBalloonColor: "#e26a6a",
    categoryBalloonAlpha: 0.8,
    valueBalloonsEnabled: false
},
legend: {
    bulletType: "round",
    equalWidths: false,
    valueWidth: 120,
    useGraphSettings: true,
    color: "#6c7b88"
}
});
},

initAmChart2: function() {
    if (typeof(AmCharts) === 'undefined' || $('#dashboard_amchart_2').size()
    === 0) {
```

Hospital Management System

```
    return;
  }

var map = AmCharts.makeChart("dashboard_amchart_2", {
  type: "map",
  "theme": "light",
  pathToImages: "../assets/global/plugins/amcharts/ammap/images/",

  dataProvider: {
    map: "worldLow",
    linkToObject: "london",
    images: [{
      id: "london",
      color: "#009dc7",
      svgPath: targetSVG,
      title: "London",
      latitude: 51.5002,
      longitude: -0.1262,
      scale: 1.5,
      zoomLevel: 2.74,
      zoomLongitude: -20.1341,
      zoomLatitude: 49.1712,

      lines: [{
        latitudes: [51.5002, 50.4422],
```

Hospital Management System

```
longitudes: [-0.1262, 30.5367]
}, {
  latitudes: [51.5002, 46.9480],
  longitudes: [-0.1262, 7.4481]
}, {
  latitudes: [51.5002, 59.3328],
  longitudes: [-0.1262, 18.0645]
}, {
  latitudes: [51.5002, 40.4167],
  longitudes: [-0.1262, -3.7033]
}, {
  latitudes: [51.5002, 46.0514],
  longitudes: [-0.1262, 14.5060]
}, {
  latitudes: [51.5002, 48.2116],
  longitudes: [-0.1262, 17.1547]
}, {
  latitudes: [51.5002, 44.8048],
  longitudes: [-0.1262, 20.4781]
}, {
  latitudes: [51.5002, 55.7558],
  longitudes: [-0.1262, 37.6176]
}, {
  latitudes: [51.5002, 38.7072],
  longitudes: [-0.1262, -9.1355]
```

Hospital Management System

```
}, {  
    latitudes: [51.5002, 54.6896],  
    longitudes: [-0.1262, 25.2799]  
}, {  
    latitudes: [51.5002, 64.1353],  
    longitudes: [-0.1262, -21.8952]  
}, {  
    latitudes: [51.5002, 40.4300],  
    longitudes: [-0.1262, -74.0000]  
}],  
  
images: [{  
    label: "Flights from London",  
    svgPath: planeSVG,  
    left: 100,  
    top: 45,  
    labelShiftY: 5,  
    color: "#d93d5e",  
    labelColor: "#d93d5e",  
    labelRollOverColor: "#d93d5e",  
    labelFontSize: 20  
}, {  
    label: "show flights from Vilnius",  
    left: 106,  
    top: 70,
```

Hospital Management System

```
    labelColor: "#6c7b88",  
    labelRollOverColor: "#d93d5e",  
    labelFontSize: 11,  
    linkToObject: "vilnius"  
  }]  
},
```

```
{  
  id: "vilnius",  
  color: "#009dc7",  
  svgPath: targetSVG,  
  title: "Vilnius",  
  latitude: 54.6896,  
  longitude: 25.2799,  
  scale: 1.5,  
  zoomLevel: 4.92,  
  zoomLongitude: 15.4492,  
  zoomLatitude: 50.2631,  
  
  lines: [{  
    latitudes: [54.6896, 50.8371],  
    longitudes: [25.2799, 4.3676]  
  }, {  
    latitudes: [54.6896, 59.9138],  
    longitudes: [25.2799, 10.7387]
```

Hospital Management System

```
}, {  
    latitudes: [54.6896, 40.4167],  
    longitudes: [25.2799, -3.7033]  
}, {  
    latitudes: [54.6896, 50.0878],  
    longitudes: [25.2799, 14.4205]  
}, {  
    latitudes: [54.6896, 48.2116],  
    longitudes: [25.2799, 17.1547]  
}, {  
    latitudes: [54.6896, 44.8048],  
    longitudes: [25.2799, 20.4781]  
}, {  
    latitudes: [54.6896, 55.7558],  
    longitudes: [25.2799, 37.6176]  
}, {  
    latitudes: [54.6896, 37.9792],  
    longitudes: [25.2799, 23.7166]  
}, {  
    latitudes: [54.6896, 54.6896],  
    longitudes: [25.2799, 25.2799]  
}, {  
    latitudes: [54.6896, 51.5002],  
    longitudes: [25.2799, -0.1262]  
}, {
```

Hospital Management System

```
latitudes: [54.6896, 53.3441],  
longitudes: [25.2799, -6.2675]  
}],
```

```
zoomControl: {  
  buttonFillColor: "#dddddd"  
},
```

```
areasSettings: {  
  unlistedAreasColor: "#15A892"  
},
```

```
imagesSettings: {  
  color: "#d93d5e",  
  rollOverColor: "#d93d5e",  
  selectedColor: "#009dc7"  
},
```

```
linesSettings: {  
  color: "#d93d5e",  
  alpha: 0.4  
},
```

```
backgroundZoomsToTop: true,
```

Hospital Management System

```
linesAboveImages: true,

"export": {
  "enabled": true,
  "libs": {
    "path": "http://www.amcharts.com/lib/3/plugins/export/libs/"
  }
}
});
},

initAmChart3: function() {
  if (typeof(AmCharts) === 'undefined' || $('#dashboard_amchart_3').size()
=== 0) {
    return;
  }

  var chart = AmCharts.makeChart("dashboard_amchart_3", {
    "type": "serial",
    "addClassNames": true,
    "theme": "light",
    "path": "../assets/global/plugins/amcharts/ammap/images/",
    "autoMargins": false,
    "marginLeft": 30,
    "marginRight": 8,
    "marginTop": 10,
```

Hospital Management System

```
"marginBottom": 26,  
"balloon": {  
  "adjustBorderColor": false,  
  "horizontalPadding": 10,  
  "verticalPadding": 8,  
  "color": "#ffffff"  
},
```

```
"dataProvider": [{  
  "year": 2009,  
  "income": 23.5,  
  "expenses": 21.1  
}, {  
  "year": 2010,  
  "income": 26.2,  
  "expenses": 30.5  
}, {  
  "year": 2011,  
  "income": 30.1,  
  "expenses": 34.9  
}, {  
  "year": 2012,  
  "income": 29.5,  
  "expenses": 31.1  
}, {
```

Hospital Management System

```
"year": 2013,
"income": 30.6,
"expenses": 28.2,
}, {
  "year": 2014,
  "income": 34.1,
  "expenses": 32.9,
  "dashLengthColumn": 5,
  "alpha": 0.2,
  "additional": "(projection)"
}],
"valueAxes": [{
  "axisAlpha": 0,
  "position": "left"
}],
"startDuration": 1,
"graphs": [{
  "alphaField": "alpha",
  "balloonText": "<span style='font-size:12px;'>[[title]] in  
[[category]]:<br><span style='font-size:20px;'>[[value]]</span>  
[[additional]]</span>",
  "fillAlphas": 1,
  "title": "Income",
  "type": "column",
  "valueField": "income",
  "dashLengthField": "dashLengthColumn"
```

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```
    }, {
      "id": "graph2",
      "balloonText": "<span style='font-size:12px;'>[[title]] in  
[[category]]:<br><span style='font-size:20px;'>[[value]]</span>  
[[additional]]</span>",
      "bullet": "round",
      "lineThickness": 3,
      "bulletSize": 7,
      "bulletBorderAlpha": 1,
      "bulletColor": "#FFFFFF",
      "useLineColorForBulletBorder": true,
      "bulletBorderThickness": 3,
      "fillAlphas": 0,
      "lineAlpha": 1,
      "title": "Expenses",
      "valueField": "expenses"
    }],
    "categoryField": "year",
    "categoryAxis": {
      "gridPosition": "start",
      "axisAlpha": 0,
      "tickLength": 0
    },
    "export": {
      "enabled": true
    }
  }
```

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```
});  
},  
  
initAmChart4: function() {  
    if (typeof(AmCharts) === 'undefined' || $('#dashboard_amchart_4').size()  
=== 0) {  
        return;  
    }  
    "valueField": "value",  
    "titleField": "country",  
    "outlineAlpha": 0.4,  
    "depth3D": 15,  
    "balloonText": "[[title]]<br><span style='font-  
size:14px'><b>[[value]]</b> ([[percents]]%)</span>",  
    "angle": 30,  
    "export": {  
        "enabled": true  
    }  
});  
  
jQuery('.chart-input').off().on('input change', function() {  
    var property = jQuery(this).data('property');  
    var target = chart;  
    var value = Number(this.value);  
    chart.startDuration = 0;
```

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```
    if (property == 'innerRadius') {  
        value += "%";  
    }  
  
    target[property] = value;  
    chart.validateNow();  
});  
,  
  
initWorldMapStats: function() {  
    if ($('#mapplic').size() === 0) {  
        return;  
    }  
  
    $('#mapplic').mapplic({  
        source: '../assets/global/plugins/mapplic/world.json',  
        height: 265,  
        animate: false,  
        sidebar: false,  
        minimap: false,  
        locations: true,  
        deeplinking: true,  
        fullscreen: false,  
        hovertip: true,
```

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```
zoombuttons: false,  
clearbutton: false,  
developer: false,  
maxscale: 2,  
skin: 'mapplic-dark',  
zoom: true  
});
```

```
$("#widget_sparkline_bar").sparkline([8, 7, 9, 8.5, 8, 8.2, 8, 8.5, 9, 8, 9], {  
  type: 'bar',  
  width: '100',  
  barWidth: 5,  
  height: '30',  
  barColor: '#4db3a4',  
  negBarColor: '#e02222'  
});
```

```
$("#widget_sparkline_bar2").sparkline([8, 7, 9, 8.5, 8, 8.2, 8, 8.5, 9, 8, 9], {  
  type: 'bar',  
  width: '100',  
  barWidth: 5,  
  height: '30',  
  barColor: '#f36a5a',  
  negBarColor: '#e02222'  
});
```

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```
$("#widget_sparkline_bar3").sparkline([8, 7, 9, 8.5, 8, 8.2, 8, 8.5, 9, 8, 9], {
    type: 'bar',
    width: '100',
    barWidth: 5,
    height: '30',
    barColor: '#5b9bd1',
    negBarColor: '#e02222'
});

$("#widget_sparkline_bar4").sparkline([8, 7, 9, 8.5, 8, 8.2, 8, 8.5, 9, 8, 9], {
    type: 'bar',
    width: '100',
    barWidth: 5,
    height: '30',
    barColor: '#9a7caf',
    negBarColor: '#e02222'
});
},
init: function() {
    this.initJQVMAP();
    this.initCalendar();
    this.initCharts();
    this.initEasyPieCharts();
    this.initSparklineCharts();
}
```

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```
this.initChat();  
this.initDashboardDaterange();  
this.initMorisCharts();  
  
this.initAmChart1();  
this.initAmChart2();  
this.initAmChart3();  
this.initAmChart4();  
  
this.initWorldMapStats();  
}  
};  
}();  
if (App.isAngularJsApp() == false) {  
    jQuery(document).ready(function() {  
        Dashboard.init(); // init metronic core componets  
    });  
}
```

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8 Database:

DATABASE : Admin

PURPOSE : Admin configuration & audit database

TECHNOLOGY: MongoDB

```
/* Switch to Admin database */
```

```
use Admin;
```

1. MongoDB_AuditLog Collection

```
db.createCollection("MongoDB_AuditLog");
```

```
db.MongoDB_AuditLog.createIndex({ auditDate: 1 });
```

2. Admin_DatabaseLog Collection

```
db.createCollection("Admin_DatabaseLog");
```

```
/* Index for database name */
```

```
db.Admin_DatabaseLog.createIndex({ databaseName: 1 });
```

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3. Admin_Parameters Collection

```
db.createCollection("Admin_Parameters");

db.Admin_Parameters.createIndex(

  { parameterGroupName: 1, parameterName: 1 },

  { unique: true }

);
```

4. Insert Default Parameters

```
db.Admin_Parameters.insertMany([

  {

    parameterGroupName: "Admin",

    parameterName: "DbBackupFolderPath",

    parameterValue: "C:/DanpheHealthInc_PvtLtd_Files/Data/DbBackup/",

    valueDataType: "string",

    description: "Database backup folder path",

    createdOn: new Date()

  },
```

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```
{  
  parameterGroupName: "Admin",  
  parameterName: "DbBackupDays",  
  parameterValue: "5",  
  valueDataType: "string",  
  description: "Number of days to keep database backup files",  
  createdOn: new Date()  
},
```

```
{  
  parameterGroupName: "Admin",  
  parameterName: "DatabaseCurrentVersion",  
  parameterValue: "2.0",  
  valueDataType: "string",  
  description: "Current database version",  
  createdOn: new Date()  
},
```

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```
{  
  parameterGroupName: "Admin",  
  parameterName: "DailyDbBackupLimit",  
  parameterValue: "5",  
  valueDataType: "string",  
  description: "Daily database backup limit",  
  createdOn: new Date()  
},
```

```
{  
  parameterGroupName: "Admin",  
  parameterName: "LiveDatabaseName",  
  parameterValue: "Danphe_EMR_LIVE",  
  valueDataType: "string",  
  description: "Live database name",  
  createdOn: new Date()  
},
```

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```
{  
  parameterGroupName: "Common",  
  parameterName: "MongoAuditFilePath",  
  parameterValue: "C:/DanpheHealthInc_PvtLtd_Files/Data/MongoAudit/",  
  valueDataType: "string",  
  description: "MongoDB audit log file path",  
  createdOn: new Date()  
}  
]);
```

5. Sample MongoDB Audit Log Insert

```
db.MongoDB_AuditLog.insertOne({  
  serverName: "SERVER01",  
  userName: "admin",  
  authenticationType: "MongoDB",  
  databaseName: "Danphe_EMR_LIVE",  
  readAccess: true,  
  writeAccess: true,  
  adminAccess: true,  
  auditDate: new Date()
```

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```
});
```

6. Stored Procedure Replacement Query

Fetch audit logs between dates and by authentication type

```
db.MongoDB_AuditLog.find({  
  auditDate: {  
    $gte: ISODate("2026-01-01"),  
    $lte: ISODate("2026-12-31")  
  },  
  authenticationType: "MongoDB"  
});
```

```
function clearCollection(name) {  
  if (db.getCollectionNames().includes(name)) {  
    db[name].deleteMany({});  
    print("Cleared collection:", name);  
  } else {  
    print("Collection not found:", name);  
  }  
}
```

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/*Clear Accounting Collections

clearCollection("ACC_Bill_LedgerMapping");

clearCollection("ACC_FiscalYear_Log");

clearCollection("ACC_InvoiceData");

clearCollection("ACC_LedgerBalanceHistory");

clearCollection("ACC_Log_EditVoucher");

clearCollection("ACC_Map_TxnItemCostCenterItem");

clearCollection("ACC_ReverseTransaction");

clearCollection("ACC_Transaction_History");

clearCollection("ACC_TransactionItems");

clearCollection("ACC_Transactions");

*/ Clear Admission & Patient Collections

clearCollection("ADT_PatientAdmission");

clearCollection("ADT_PatientBedInfo");

clearCollection("PAT_Patient");

clearCollection("PAT_PatientVisits");

clearCollection("PAT_PatientFiles");

clearCollection("PAT_PatientInsuranceInfo");

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/* Clear Billing Collections

clearCollection("BIL_BillingTransaction");

clearCollection("BIL_InvoiceReturn");

clearCollection("BIL_InvoiceReturnItems");

clearCollection("BIL_Deposit");

clearCollection("BIL_CashHandover");

/* Clear Clinical Collections

clearCollection("CLN_PatientNotes");

clearCollection("CLN_Diagnosis");

clearCollection("CLN_Allergies");

clearCollection("CLN_MedicationPrescription");

/* Clear Pharmacy Collections

clearCollection("PHRM_Stock");

clearCollection("PHRM_StockTxnItems");

clearCollection("PHRM_Invoice");

clearCollection("PHRM_InvoiceItems");

clearCollection("PHRM_Prescription");

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/* Clear Lab & Radiology Collections

clearCollection("LAB_TestRequisition");

clearCollection("LAB_LabReports");

clearCollection("RAD_PatientImagingReport");

/* Clear Ward & OT Collections

clearCollection("WARD_Stock");

clearCollection("WARD_Consumption");

clearCollection("OT_BookingDetails");

/* Clear Logs & Audit Collections

clearCollection("MongoDB_AuditLog");

clearCollection("LoginInformation");

9 Advantages

1. Reduces Manual Work and Paperwork

The Hospital Management System significantly reduces the dependence on manual, paper-based processes. HMS such as patient registration, appointment scheduling, billing, and report generation are automated, minimizing the need for physical files and registers. This not only saves time and effort for hospital staff but also reduces storage requirements and the risk of losing or damaging important documents

2. Improves Accuracy and Efficiency

By automating data entry and processing, the system minimizes human errors commonly associated with manual record-keeping, such as incorrect entries or miscalculations in billing. Standardized forms and validation checks ensure data consistency and correctness.

3. Easy Access to Patient Information

The system provides quick and secure access to patient records, including medical history, prescriptions, lab reports, and treatment details. Authorized users such as doctors and nurses can retrieve patient information instantly from a centralized database, enabling faster diagnosis and better decision-making.

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4. Better Resource Management

A Hospital Management System helps in effective management of hospital resources such as doctors, staff, beds, medical equipment, and medicines. Features like scheduling, inventory tracking, and usage reports allow administrators to monitor resource utilization and avoid shortages or overuse.

5. Enhanced Patient Satisfaction

With faster registration, reduced waiting times, accurate billing, and improved communication, patients experience smoother and more transparent services. Online appointment booking, timely notifications, and easy access to medical reports increase convenience for patients. Overall, efficient hospital operations and better-quality care lead to higher patient trust and satisfaction.

10 Limitations

1. High Initial Setup Cost

The implementation of a Hospital Management System requires significant initial investment in hardware, software, and infrastructure. Small hospitals and clinics may find it financially challenging to adopt the system due to costs related to servers, licensed software, and system configuration.

2. Training and Adaptation Challenges

Hospital staff such as doctors, nurses, and administrative personnel may not be familiar with computerized systems. Proper training is required to ensure effective usage of the system, and during the initial phase, productivity may decrease as users adapt to the new workflow.

3. Dependence on Technology

The system is highly dependent on computer hardware, network connectivity, and power supply. Any technical failure such as server downtime, network issues, or power outages can disrupt hospital operations and delay patient care.

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4. Data Security and Privacy Risks

Hospital Management Systems store sensitive patient information, making them potential targets for cyberattacks. Unauthorized access, data breaches, or system vulnerabilities can compromise patient confidentiality and violate data protection regulations.

5. Limited Customization

The system may not fully support the specific operational requirements of all hospitals. Customizing the software to meet unique departmental workflows or policies can be complex, time-consuming, and costly.

6. Data Migration Issues

Converting existing paper-based or legacy digital records into the new system is a challenging process. Data loss, inconsistencies, or errors may occur during migration, which can affect the accuracy of patient records.

7. Maintenance and Upgrade Requirements

Regular system maintenance, updates, and bug fixes are required to keep the system secure and efficient. This may lead to additional operational costs and temporary service interruptions during system upgrades.

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8. Scalability Constraints

As the hospital grows, the system may face performance issues if it is not designed to handle increased data volume and user load. Scaling the system may require infrastructure upgrades and technical expertise.

9. User Dependency and Human Errors

Although the system reduces manual errors, incorrect data entry by users can still lead to inaccurate records, billing mistakes, or treatment delays. The system's effectiveness depends heavily on correct and timely data input.

10. Legal and Compliance Challenges

Hospitals must comply with healthcare regulations and data protection laws. The system may require continuous updates to align with changing legal standards, and failure to comply can result in legal penalties.

11 Future Enhancements

Every project has certain limitations due to technological, time, and resource constraints. *Hospital Management System* is a reliable and efficient solution for managing daily hms, but there are a few areas where improvements can be made to increase its overall performance, scalability, and user experience.

To make HMS more robust and user-friendly, several improvements can be introduced in future versions:

1. Mobile Application Integration

In the future, the Hospital Management System can be extended with mobile applications for Android and iOS platforms. This will allow doctors to access patient records remotely, patients to book appointments, view reports, and receive notifications, and hospital staff to manage hms more efficiently on the go.

2. Telemedicine and Online Consultation

The system can be enhanced to support telemedicine features such as video consultations, online prescriptions, and virtual follow-ups. This will be especially beneficial for patients in remote areas and for reducing hospital overcrowding while ensuring continuous medical care.

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3. Artificial Intelligence and Machine Learning Integration

AI and ML algorithms can be incorporated to assist in disease prediction, diagnosis support, and treatment recommendations. The system can analyze patient history, lab reports, and medical trends to provide decision support to doctors and improve clinical outcomes.

4. Cloud-Based Deployment

Migrating the system to a cloud platform will improve scalability, availability, and data backup. Cloud-based HMS can enable access from multiple locations, reduce infrastructure costs, and ensure disaster recovery through automated backups and redundancy.

5. Integration with Wearable and IoT Devices

Future versions can integrate with wearable devices and IoT-enabled medical equipment to collect real-time patient health data such as heart rate, blood pressure, glucose levels, and oxygen saturation. This will support continuous monitoring and early detection of critical conditions.

6. Advanced Data Analytics and Reporting

The system can be enhanced with advanced analytics dashboards to provide insights into hospital performance, patient flow, resource utilization, and financial trends. Predictive analytics can help hospital management in planning staff allocation and inventory management.

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7. Electronic Health Record (EHR) Interoperability

Future enhancements may include interoperability with national or regional Electronic Health Record systems. This will allow secure data sharing between hospitals, clinics, laboratories, and insurance providers, ensuring continuity of patient care across healthcare facilities.

8. Automated Appointment and Queue Management

The system can be improved with smart scheduling algorithms to optimize appointment slots and reduce patient waiting time. Automated queue management using tokens, SMS alerts, and display systems can enhance patient experience.

9. Enhanced Security and Blockchain Integration

Advanced security mechanisms such as biometric authentication, multi-factor authentication, and blockchain-based data storage can be implemented to ensure data integrity, transparency, and protection against unauthorized access.

10. Insurance and Billing Automation

Integration with insurance providers can automate claim processing, policy verification, and reimbursement tracking. This will reduce manual work, minimize billing errors, and speed up financial settlements.

12 Conclusion

The Hospital Management System represents a significant step toward the digital transformation of healthcare administration by providing an integrated platform that streamlines and automates hospital operations.

Throughout the development and implementation of the system, key hospital processes such as patient registration, appointment scheduling, electronic medical record management, laboratory services, pharmacy management, billing, and staff administration have been effectively computerized.

This integration minimizes manual effort, reduces paperwork, and ensures that accurate and up-to-date information is readily available to authorized users.

By centralizing data within a secure database, the system enhances data consistency, integrity, and confidentiality. It reduces the likelihood of human errors associated with manual record-keeping while improving the speed and efficiency of data retrieval.

Doctors and medical staff benefit from timely access to patient histories, test results, and treatment records, enabling better clinical decision-making and continuity of care. Administrative staff experience improved workflow management, reduced operational delays, and greater transparency in billing and reporting processes.

The Hospital Management System also supports improved resource utilization by enabling better tracking of hospital assets, staff schedules, and service availability.

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Automated reporting and analytics provide hospital management with valuable insights into operational performance, patient flow, and financial status, supporting informed decision-making and strategic planning.

Additionally, role-based access control and security mechanisms ensure that sensitive patient information is protected in compliance with healthcare data standards.

Overall, the system enhances the quality of healthcare services by improving operational efficiency, reducing costs, and increasing patient satisfaction. It provides a scalable and flexible foundation that can be expanded in the future to include advanced features such as online patient portals, telemedicine integration, mobile access, and data analytics.

In conclusion, the Hospital Management System serves as a reliable, efficient, and modern solution that addresses the growing demands of healthcare institutions and contributes to the delivery of safe, timely, and high-quality patient care.

13 Bibliography

The development of **Hospital Management System** was inspired by several online resources, official documentation, and research papers that contributed to the understanding of Node.js, MongoDB, and CRUD operations. This bibliography lists the references and resources used during the design and implementation of the project.

Node.js Documentation – The official Node.js documentation (<https://nodejs.org/en/docs/>) was a primary reference for understanding the fundamentals of Node.js, including modules, asynchronous programming, event-driven architecture, and HTTP server creation. It provided the base knowledge required to set up the server-side logic for the project and build efficient RESTful APIs for CRUD operations.

MongoDB Documentation – The official MongoDB documentation was used extensively to understand database design, schema modeling using Mongoose, and performing operations such as Create, Read, Update, and Delete. This helped in managing hms efficiently within the database and ensuring scalability of the system.

Express.js Guide – The Express.js official website (<https://expressjs.com/>) was referred to for building robust APIs and middleware integration. Express was used as the backend framework to handle routes, middleware, and HTTP requests efficiently.

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W3Schools and GeeksforGeeks – Tutorials and articles from W3Schools (<https://www.w3schools.com/>) and GeeksforGeeks (www.geeksforgeeks.org/) were used to clarify specific concepts of JavaScript, Node.js, and REST APIs. These platforms provided examples and explanations that were instrumental in implementing user authentication and request handling.

Stack Overflow – Community discussions and solutions from Stack Overflow played a major role in debugging errors and finding best practices for connecting Node.js applications with MongoDB, handling asynchronous calls, and structuring project directories.

NPM and Open-Source Packages – The Node Package Manager repository was used to explore and install dependencies like Mongoose, Express, Body-Parser, and Nodemon. Documentation for each package was reviewed to understand proper implementation and configuration.

Research Articles and Blogs – Articles on modern hms management systems and productivity tools were referred to, which provided insights into features like categorizing hms, setting deadlines, and improving user experience.

Hospital Management System

14 References

- I am thankful to my guide **Dr. Jyotsna Salet** for guiding me. I am also thankful to whole staff of the Computer Department for giving me huge support for my project.

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